

Design and Implementation of an Intelligent Pastry Manufacturing Cell Using PLC Coordination, Computer Vision, Collaborative Robotics, IoT Monitoring, and Cybersecurity

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Abstract — This paper presents the design and physical implementation of an automated manufacturing cell integrating machining, industrial automation, computer vision, collaborative robotics, IoT monitoring, and cybersecurity. A stepped cylindrical workpiece was manufactured from Aluminum 6061 on a ROMI T240 lathe, serving as both a production component and a classification reference for the vision system. Electrical consumption during machining was monitored non-invasively using an SCT-013 current transformer, an ADS1115 16-bit ADC, and an ESP32 gateway, which continuously estimated RMS current, active three-phase power, accumulated energy consumption, operating cost, and discrete machine states (OFF, IDLE, CUTTING, FAULT). Data was transmitted via MQTT to a Node-RED dashboard, enabling remote real-time visualization and historical logging. Cell automation was integrated in two Siemens S7-1200 PLCs interconnected via an S7 network, exchanging 96 bytes of bidirectional process data. A Cognex industrial vision sensor classified three reference objects into product recipes, dynamically configuring flavor-dependent processing times through PLC multiplexor logic. Material handling, product assembly, and palletizing were performed by a UFACTORY xArm Lite 6 collaborative robot programmed in Python using a priority-based Finite State Machine capable of managing two concurrent workpieces. Performance was evaluated through machine utilization, OEE, cycle time, production rate, energy distribution per operating state, and estimated operating cost, yielding an OEE of 7% consistent with single-unit job production conditions.

Keywords — Automated Manufacturing Cell, Collaborative Robotics, Computer Vision, Programmable Logic Controller (PLC), Industrial Internet of Things (IIoT), Finite State Machine (FSM), Overall Equipment Effectiveness (OEE), Energy Monitoring

I. INTRODUCTION

A. State of the Art

The emergence of the fourth industrial revolution, broadly termed Industry 4.0, has introduced two converging forces reshaping modern manufacturing: an application-pull driven by demands for individualization, flexibility, and resource efficiency, and a technology-push characterized by increasing mechanization, digitalization, and networking of production systems (Lasi et al., 2014). At the core of this paradigm lies the Smart Factory concept, described as manufacturing environments “completely equipped with sensors, actors, and autonomous systems” (Lasi et al., 2014, p. 240). Within these environments, cyber-physical systems merge physical processes with their digital representations

to enable real-time monitoring, control, and self-optimization.

Within this framework, collaborative robots have emerged as key enablers of flexible industrial automation. According to Villani et al. (2018), cobots provide a balance between robotic precision and human adaptability by integrating high accuracy, repeatability, and speed with the flexibility and cognitive capabilities of human workers. Their adoption aligns with Industry 4.0 objectives because collaborative automation seeks improvements in efficiency, productivity, and cost reduction while supporting greater flexibility and interoperability across production systems (Villani et al., 2018). Furthermore, the authors report increasing implementation of cobots in small and medium-sized enterprises due to the growing availability of affordable and user-friendly solutions, making them particularly relevant for didactic manufacturing environments.

Industrial machine vision systems constitute another foundational technology in automated manufacturing. Traditional visual inspection methods are often insufficient for modern production requirements because they rely on subjective human judgment and produce inconsistent results. As noted by Czimmermann et al. (2020), traditional inspection remains “a non-measurable process with variable and subjective outcomes” (para. 3), motivating the development of automated defect detection systems. Beyond quality control applications, industrial vision systems are increasingly employed for object identification and classification tasks that dynamically influence downstream manufacturing processes, enabling greater production flexibility and reducing manual intervention.

For real-time process monitoring, the MQTT protocol has established itself as one of the dominant communication standards in Industrial Internet of Things (IIoT) environments. Mishra and Kertesz (2020) attribute its widespread adoption to characteristics such as low overhead, simplicity, and ease of deployment, making it particularly suitable for constrained environments. The protocol was specifically designed to minimize bandwidth consumption and device resource requirements while maintaining reliable communication performance (Mishra & Kertesz, 2020). Their analysis further reports an average annual growth rate exceeding 32% in MQTT-related publications between 2004 and 2019, with applications spanning healthcare, agriculture, logistics, smart cities, and

industrial manufacturing environments. Through its publish-subscribe architecture operating over TCP/IP, MQTT enables modular and scalable communication infrastructures that are particularly suitable for heterogeneous IIoT systems requiring integration among multiple sensors, controllers, and monitoring platforms.

B. Challenge Description

In this work, a didactic manufacturing cell was automated and monitored to simulate an industrial pastry production facility. The cell is governed by two Siemens Simatic S7-1200 PLCs, each paired with a 7-inch Simatic HMI, and incorporates electropneumatic and electronic actuators, optoelectronic sensors, a UFACTORY xArm Lite 6 collaborative robot, and a Cognex industrial vision sensor. Complementary hardware including a pneumatic funnel, raw material supply grid, palletizing cage, and RFID access control module was designed and fabricated to achieve complete process integration. The cell operates across two sequential stages: a pneumatic processing stage and a machining simulation stage, each controlled independently while maintaining communication through an S7 network.

Concurrently, a physical aluminum workpiece was manufactured on a ROMI T240 manual lathe at the CIMA machining facility. This component, together with a milled aluminum piece and a metal hook, constitutes the set of reference objects used by the Cognex vision system to classify incoming product types at the beginning of each production cycle. Each classified object corresponds to a unique production recipe within the simulation: the lathe-machined component represents Chocolate, the milled component Strawberry, and the metal hook Vanilla.

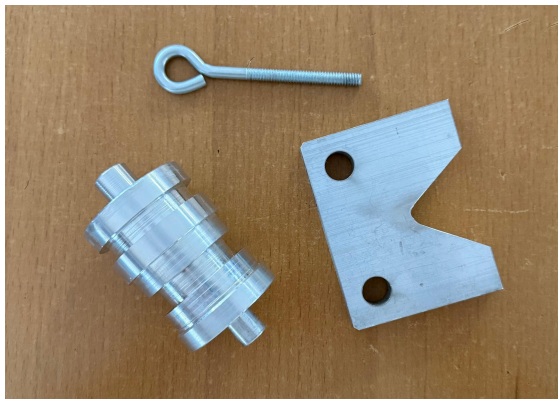


Figure 1. Pieces Used for the Classification System

Classification outputs are transmitted digitally to the PLCs, which dynamically adjust processing times, actuator sequences, and cobot routines accordingly. The collaborative robot performs raw material feeding, inter-conveyor transfer, product assembly, and palletizing tasks. Access to HMI interfaces is protected through RFID-based authentication connected to the PLC via relay logic, providing operator-level cybersecurity controls. Electrical consumption during machining is monitored in real time using an SCT-013 current transformer, an ADS1115 analog-to-digital converter, and an ESP32 gateway, with collected data transmitted through MQTT and

visualized using a Node-RED dashboard accessible through the institutional network.

C. Team's Proposal

To address the manufacturing and automation challenge, the team proposed and physically implemented an integrated mechatronic system designed to simulate an automated cake manufacturing facility. The technical approach bridges physical machining operations with advanced industrial automation, utilizing a dual-PLC architecture, collaborative robotics, computer vision, and Industrial Internet of Things (IIoT) technologies. The specific solution is divided into three core technical pillars:

a. Plant 1: Custom Machining and Non-Invasive IoT Instrumentation

For the machining stage at CIMA, a cylindrical stepped workpiece was designed and manufactured from Aluminum 6061 using a ROMI T240 lathe. To monitor this process, a dedicated, non-invasive IoT instrumentation system was developed. This system utilizes an SCT-013 current transformer and an ESP32 microcontroller paired with an ADS1115 16-bit ADC to acquire high-resolution electrical data. An embedded algorithm processes these signals to estimate active three-phase power, accumulated energy consumption, and discrete machine states (OFF, IDLE, CUTTING, and FAULT). This data is transmitted in real-time via the lightweight MQTT protocol to a centralized Node-RED dashboard, enabling continuous remote monitoring and historical performance analysis.

b. Plant 2: Advanced Automation, Vision, and Collaborative Robotics

The automation of Cell 3105 required using two Siemens S7-1200 PLCs interconnected via an S7 communication network. A Cognex industrial vision sensor was integrated as a primary quality gate, classifying pieces into three distinct categories mapped to the simulation theme: the lathe-machined piece (Chocolate), the milled piece (Strawberry), and the metal hook (Vanilla). The PLC control logic employs dynamic array indexing to track the specific ID of each part as it moves through the line, allowing multiplexors to assign precise, flavor-dependent processing times for the drilling and mixing stations.

Furthermore, the UFACTORY xArm Lite 6 collaborative robot was programmed using Python 3 and the native xArm-Python-SDK, bypassing the limitations of rigid, linear Blockly programming. This approach implemented a pseudo-concurrent Finite State Machine (FSM) with priority-based task scheduling. This architecture allows the cobot to manage up to two products simultaneously through distinct phases: base pick-and-place, inter-conveyor transfer synchronization, topping assembly, and a structured two-level palletizing sequence that includes the automated placement of separator lids.

c. System Innovation and Cybersecurity Integration

To elevate the baseline project requirements, several innovative modules were integrated. An RFID-based access control system was implemented and wired to the PLC via relays to enforce cybersecurity protocols, ensuring only

authorized users can access and modify the HMI interfaces. Additionally, custom end-effectors (grippers) were designed and fabricated for the cobot to reliably handle the specific geometries of the raw material grid, the pneumatic funnel, and the final assembled product. Finally, a unified monitoring architecture was developed, enabling real-time dashboard visualization of both Plant 1 and Plant 2 operations from any network location within the campus.

II. TECHNOLOGICAL DEVELOPMENT AND IMPLEMENTATION

A. Plant 1 - CIMA: Machining & IoT Instrumentation

a. Machining Process

The technical drawing of the proposed component is presented in Figure 2. The manufacturing challenge consisted of the development of a cylindrical stepped workpiece featuring multiple diameters and section lengths. The dimensional specifications were defined based on machining feasibility and system integration requirements. Additionally, coordination was carried out with the MCK team during the design phase to ensure geometric compatibility between the manufactured workpiece and the V-shaped support fixtures developed independently by that team. As a result, the selected diameters ensured proper alignment and handling of the component during subsequent stages. The component was manufactured using a conventional ROMI T240 lathe. The workpiece material consisted of Aluminum 6061, supplied as a cylindrical bar with an initial diameter of 2 in and a length of 7 in.

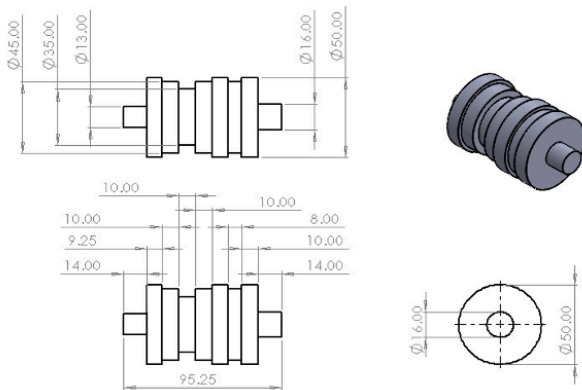


Figure 2. Technical Drawing of the Workpiece

Prior to machining, the raw material was sectioned using a horizontal band saw (Figure 3). An additional allowance of approximately 80 mm was intentionally left on the cut stock to prevent interference between the lathe guard and the final workpiece geometry, as well as to have enough work material in the chuck and prevent this from moving or going out due to the spindle velocity.



Figure 3. Raw Material Cutting Operation Using the JET Horizontal Band Saw

The workpiece was secured in the lathe chuck (Figure 4), and an initial facing operation was performed to establish a reference surface and obtain the required external diameter of 50 mm, as shown in Figure 5. Once the reference geometry had been established, a series of roughing passes was carried out to remove excess material and generate the stepped profile of the component.



Figure 4. Workpiece Mounted in the ROMI T240 Chuck

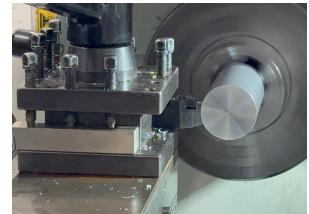


Figure 5. Initial Facing Operation

The machining process consisted primarily of external turning operations to produce the diameters specified in the design. The final geometry incorporated a maximum diameter of $\varnothing 50$ mm, intermediate diameters of $\varnothing 45$ mm and $\varnothing 35$ mm, and reduced end sections of $\varnothing 16$ mm and $\varnothing 13$ mm. Multiple geometric transitions were generated according to the dimensional requirements defined in the technical drawing. This process can be seen in Figure 6 and 7. During machining, the spindle operated at a nominal speed of 1080 RPM, providing stable cutting conditions and consistent material removal performance, as shown in Figure 8.

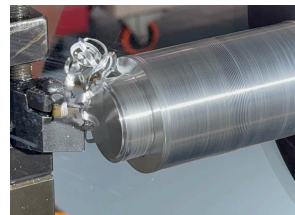


Figure 6. Rough Turning Operation for Material Removal

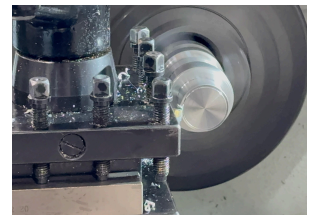


Figure 7. Machining of Stepped Diameters and Profile Features

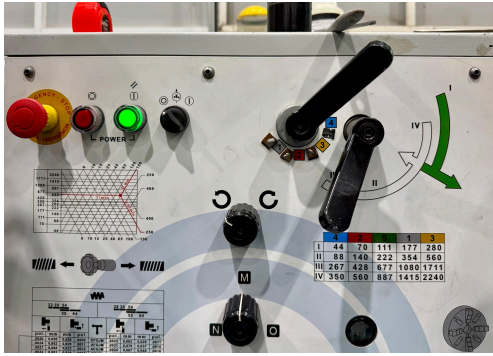


Figure 8. Spindle Speed Configuration Used During Machining (1080 RPM)

Throughout the machining process, dimensional verification was continuously performed using a Vernier caliper. Critical diameters and lengths were periodically measured following each relevant operation to ensure compliance with the specified dimensions and to minimize dimensional deviations. The machined component achieved the specified overall length of 95.25 mm and incorporated multiple stepped features with widths ranging from 8 mm to 14 mm.

The completed component is presented in Figures 9 and 10, which show the Fusion 360 model and the physical workpiece manufactured on the ROMI T240 lathe, respectively. The dimensional validation of the manufactured component ensures that the part is suitable for subsequent integration into the monitoring and data acquisition stage of the project. Since the geometry of the workpiece directly influences cutting conditions, maintaining geometric conformity with the digital model is essential for obtaining reliable process signals.

In this context, the verified agreement between the CAD model and the physical component guarantees that the machining conditions used during production correspond to the expected process parameters. This is particularly relevant for the analysis of current consumption and energy usage, as stable and repeatable cutting conditions are required to ensure that the electrical signals acquired from the ESP32-based IoT system accurately reflect the machining behavior.

Therefore, the validated manufacturing process supports the integrity of the collected electrical consumption data, enabling its use for further analysis of machining efficiency and energy performance within the production system.

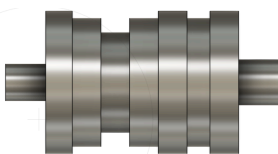


Figure 9. Fusion 360 Model of the Finished Component

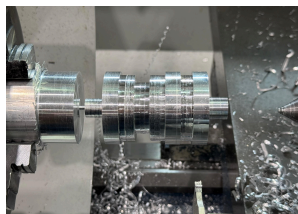


Figure 10. Manufactured Component

b. IoT Monitoring System

The electrical consumption of the ROMI T240 lathe was monitored using a non-invasive current measurement system based on an SCT-013 current sensor and an ESP32 IoT gateway.

i. Three-Phase Monitoring Enclosure

To enable safe and reliable current measurements, a dedicated monitoring enclosure was designed and assembled (Figure 11). The enclosure incorporated a three-phase industrial connector, including the three power lines and an earth conductor. All internal wiring was implemented using 12 AWG conductors to ensure adequate current-carrying capacity and compliance with electrical safety requirements. Ferrules were crimped onto all wire terminations to facilitate secure connections within the enclosure.

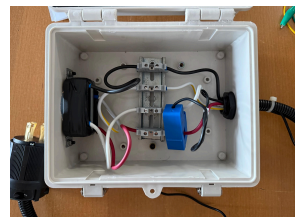


Figure 11. Inside of the Three-Phase Monitoring Enclosure



Figure 12. Installation of the Monitoring Cabinet Between the Power Supply and the ROMI T240 Lathe

The monitoring cabinet was installed between the electrical supply and the ROMI T240 lathe. This arrangement provided access to the individual phase conductors while maintaining normal machine operation. Since the SCT-013 is a non-invasive current sensor, it was clamped around one of the phase conductors without requiring any direct electrical connection to the power circuit. Consequently, the current flowing through the selected phase of the lathe could be measured safely and continuously during machining operations.

ii. Signal Conditioning and Data Acquisition Circuit

The SCT-013 current transformer produces a low-amplitude alternating voltage proportional to the current flowing through the monitored phase conductor (PowerUC Electronics, n.d.). Since the output signal contains both positive and negative half-cycles, direct acquisition by the analog-to-digital converter is not possible because the ADC operates only within the positive supply range.

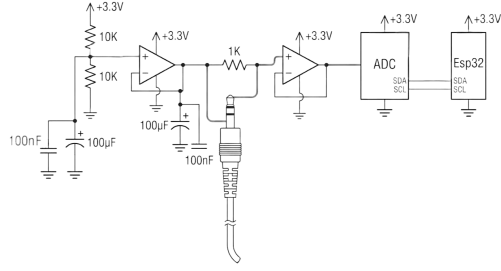


Figure 13. Electrical Schematic of the Signal Conditioning and Data Acquisition Circuit

To address this limitation, a signal conditioning stage was designed and implemented (Figure 13). The first stage consists of a voltage divider formed by two 10 kΩ resistors connected between the 3.3 V supply and ground. This arrangement generates a reference voltage of approximately 1.65 V, corresponding to half of the supply voltage.

The reference voltage is buffered using an operational amplifier configured as a voltage follower. Capacitors were incorporated around the divider network to attenuate noise and improve voltage stability.

The SCT-013 sensor output is connected across a 150 Ω burden resistor. The burden resistor converts the secondary current produced by the current transformer into a measurable voltage signal according to Ohm's law. The resulting AC voltage is superimposed on the 1.65 V virtual ground, effectively shifting the waveform into the operating range of the acquisition system.

A second operational amplifier, also configured as a voltage follower, isolates the sensor and conditioning stages from the ADC input. This buffering stage improves signal integrity, and ensures that the waveform delivered to the converter accurately represents the current measured by the SCT-013 sensor.

The conditioned signal is acquired by an ADS1115 16-bit analog-to-digital converter connected to the ESP32 through the I²C communication interface. Compared to the ESP32 internal ADC, the ADS1115 provides higher resolution and improved measurement accuracy, allowing more reliable current estimation (Admin, 2026).

iii. Current Measurement Algorithm

The ESP32 continuously acquires samples from the ADS1115 to estimate the electrical current consumed by the ROMI T240 lathe. The calculations were performed using the formulas presented by Storr and Storr (2026) in the RMS Voltage Tutorial.

For each measurement cycle, 500 samples are collected from the conditioned waveform. During the acquisition window, the algorithm determines the maximum and minimum voltage values measured by the ADC:

$$V_{pp} = V_{max} - V_{min}$$

where V_{pp} is the peak-to-peak voltage of the sampled waveform.

Assuming a sinusoidal waveform, the RMS voltage is estimated from the peak-to-peak value using:

$$V_{RMS} = 0.3536 \times V_{pp}$$

The RMS voltage is then converted into current using the burden resistor (150 Ω), as follows:

$$I_{RMS} = \frac{V_{RMS}}{150 \Omega}$$

Finally, the primary current flowing through the monitored conductor is obtained by applying the SCT-013 current transformation ratio:

$$I = I_{RMS} \times 2000$$

This method allows the electrical current consumed by the lathe to be estimated from the sampled waveform without requiring direct electrical contact with the power circuit.

iv. Power and Energy Estimation

After calculating the RMS current, the ESP32 estimates the active three-phase power using the measured current, the nominal line voltage, and an assumed power factor. Since direct power-factor measurements were not available, a power factor of 0.85 was adopted. This value is representative of typical industrial machine-tool drive systems and falls within the range commonly reported for such equipment under normal loading conditions (Schneider Electric, n.d.; Power, 2026). The active three-phase power is then calculated as follows:

$$P = \sqrt{3} \times V \times I \times PF$$

where:

- $V = 220 \text{ V}$ (line voltage)
- I is the measured current
- $PF = 0.85$ (assumed power factor)

Electrical energy consumption is calculated by integrating power over time:

$$E = \sum_{i=1}^n P_i \times \Delta t_i$$

The accumulated energy value is expressed in kilowatt-hours (kWh) and is continuously updated during machine operation.

v. Data Transmission and Visualization

After each acquisition cycle, the ESP32 processed the sampled data and generated a set of operational and energy-related indicators representative of the electrical behavior of the ROMI T240 lathe. These variables were converted into string format and transmitted using the Message Queuing Telemetry Transport (MQTT) protocol.

MQTT was selected due to its lightweight architecture, low bandwidth requirements, and suitability for Industrial Internet of Things (IIoT) applications (Simson M, 2025). Communication was implemented using a Mosquitto MQTT broker hosted on a dedicated workstation connected to the university network (Yudy's computer). This workstation

served as the central communication hub between the ESP32 IoT gateway and the monitoring platform. Operating under a publish-subscribe architecture, the broker received data published by the ESP32 and distributed the information to all subscribed clients.

The ESP32 was connected to the university wireless network and periodically published process variables to dedicated MQTT topics. A separate topic was assigned to each monitored variable, allowing modular data management and simplified dashboard integration. The transmitted variables included:

- Current (A)
- Active power consumption (kW)
- Accumulated energy consumption (kWh)
- Estimated operating cost (MXN)
- Machine operating state (OFF, IDLE, CUTTING, or FAULT)
- Accumulated OFF time (s)
- Accumulated IDLE time (s)
- Accumulated CUTTING time (s)
- Accumulated FAULT time (s)

For visualization purposes, a Node-RED dashboard was developed and configured as an MQTT subscriber. The dashboard continuously monitored the corresponding topics and updated the graphical interface whenever new measurements became available. The interface incorporated numerical indicators, trend charts, energy consumption metrics, and machine status displays, providing operators with real-time insight into machine utilization and electrical performance.

The integration of MQTT and Node-RED enabled continuous remote monitoring of the machining process while maintaining a low-cost and scalable IIoT architecture (Figure 14). By combining current measurements, power and energy calculations, and machine-state information, the dashboard provided a comprehensive overview of equipment utilization, operational efficiency, and energy consumption throughout the manufacturing process.

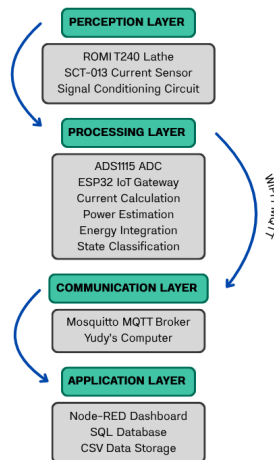


Figure 14. IoT Architecture for Real-Time Monitoring of the ROMI T240 Lathe

B. Plant 2 - Cell 3105: Automation & Robotics

a. Computer Vision Criterion (Quality Gate)

The Cognex industrial vision sensor was implemented as the primary quality gate of the manufacturing cell. Its function is to identify the type of part entering the process and communicate the classification result directly to the Siemens S7-1200 PLC through discrete digital signals.

Before each production cycle, the operator places one of three reference objects inside the Cognex inspection area: the lathe manufactured aluminum part, the milled aluminum part, or a metal hook. These objects were selected to represent the three product recipes of the pastry manufacturing simulation. The lathe part corresponds to the Chocolate recipe, the milled part corresponds to the Strawberry recipe, and the metal hook corresponds to the Vanilla recipe.

The Cognex sensor was electrically integrated into the control architecture of the Manufacturing PLC (front left) using a shared 24 VDC industrial power supply. Both the PLC and Cognex were connected to the same voltage reference and common ground (0 VDC / GND) through the control panel terminal distribution. Maintaining a common electrical reference between both devices was essential to guarantee stable digital communication, reduce electrical noise susceptibility, and avoid undefined voltage states during signal transitions (WDF,2025).

Communication between the PLC and the vision system was implemented using one digital output and three digital inputs of the PLC, as well as one digital input and three digital outputs of the Cognex sensor. PLC output Q8.7, identified in the control program as Cognex_IN0 was wired to the trigger input of the Cognex sensor, whereas the digital outputs generated by the camera were connected directly to PLC inputs I9.0, I9.1, and I9.2, corresponding to Cognex_OUT0, Cognex_OUT1, and Cognex_OUT2 respectively in the ladder program of the PLC. These outputs transmit the inspection result detected by the vision algorithm.

During operation, the PLC sequencing logic activates the trigger signal through output Q8.7 whenever the HMI designed button is pressed. Immediately after, a timer pulse (TP) instruction is activated to maintain the trigger active for one second, ensuring efficient acquisition time for stable image capture and processing. Once the image is processed internally by the Cognex sensor, the corresponding digital output is activated and sent to the PLC.

The implemented architecture was configured to support three independent classifications. Therefore, the PLC monitors the status of the three Cognex outputs (whenever the signal of the trigger is activated with the HMI button) and stores the detected result inside the Communication Data Block variable: Cognex_Code and its own local tag CognexID. The classification logic is defined as follows:

I9.0 (Cognex_OUT0) → Classification 0 (Chocolate) →
Cognex_Code = CognexID=0

- I9.1 (Cognex_OUT1) → Classification 1 (Vanilla) →
Cognex_Code = CognexID = 1
I9.2 (Cognex_OUT2) → Classification 2 (Strawberry) →
Cognex_Code = CognexID = 2

The resulting code is later used by the PLC state machine to determine the corresponding actuation or sorting sequence within the manufacturing cell.

Additionally, a fault-handling condition was incorporated into the inspection logic to improve process reliability. If none of the Cognex outputs becomes active during the acquisition window, the PLC automatically assigns the value -1 to the variable Cognex_Code. This condition represents an invalid or inconclusive inspection result and may be caused by incorrect part positioning, inadequate illumination, motion blur, or failure to detect a valid object within the camera region of interest. Under this condition, the PLC does not execute any manipulation or classification movement, preventing incorrect operations caused by unreliable image data.

Originally, the system design considered the implementation of an additional “task completed” signal from the Cognex sensor to indicate the end of image processing. However, due to a third classification proposed by the team since the beginning of the planification, the available communication resource, originally assigned to the vision sensor output Cognex_OUT2 was reassigned as an additional classification output.

Once the classification is completed, the detected ID is transmitted to the Pneumatic PLC through the PUT and GET communication network. The receiving PLC stores the value in the variable CognexID, which becomes the main decision parameter for the subsequent manufacturing operations. Indicator lights are activated to provide visual feedback of the detected product type, while the classification code is simultaneously used to select the appropriate process parameters.

A multiplexor (MUX) instruction was implemented to automatically assign different punching times according to the detected product type. Therefore, the vision system not only performs product identification but also dynamically configures the manufacturing recipe, enabling flexible production without manual intervention.

b. Control Logic and Sequencing

Recalling that a dual-PLC architecture was used for this project, a crucial step regarding the communication was to establish an S7 network that allowed the exchange of information between CPUs. This communication method involves the creation of two identical (in content) data blocks (one on each one of the PLC's) where information will be read and written in real time by using the blocks PUT (write to the other data block) and GET (read from the other data block) from the TIA portal interface.

To implement this protocol, it is important to highlight that data blocks are arranged as a list of tags, each one of them having its own associated memory direction where the

information should be written or read from. Recalling this, to reach the integration level of this project, a continuous bidirectional communication was established, where a total of 96 bytes of information are exchanged between the PLCs.

The reason behind the necessity to communicate such a great amount of information, is because two strings are shared between devices, representing 74 bytes by themselves, while the rest can be attributed to sensor states, classifications, execution codes, variables, etc.

The manufacturing process begins with the Machining PLC (front left) by setting the corresponding tags to 0, 1 or 2 depending on the classification received from the Cognex vision sensor (if the classification is invalid, the code is written in a certain way that the value -1 is never used, so it waits until receiving a valid ID). As the cobot is wired to the Pneumatic PLC (front right), the S7 network and its corresponding data blocks are used to share the classification between both CPUs.

Once the Pneumatic PLC receives the classification, it sets a byte and sends the first three bits to the cobot through its wired inputs, each different combination corresponding to the piece that is next to be manufactured. The Byte is set as follows:

- Classification 0 (Chocolate Flavor) → Decimal number
assigned = 1 → Byte = 00000001
Classification 1 (Vanilla Flavor) → Decimal number
assigned = 2 → Byte = 00000010
Classification 2 (Strawberry Flavor) → Decimal number
assigned = 3 → Byte = 00000011

Through its programming and mapped positions, the cobot knows where to move over the grid to get the corresponding raw material that will be the input for the immediate process. Right after, the cobot grabs the piece and moves it to the cell input at the pneumatic funnel.

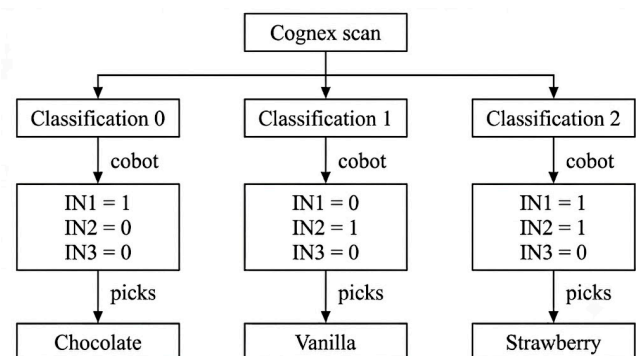


Figure 15. Flow Chart for Input Detection

Once a piece is located at the starting position of the pneumatic process and the optoelectronic sensor is covered, with the use of a normally closed contact, a timer ON (200ms) is used to delay the activation of a timer OFF that keeps the starting piston extended (400ms) pushing the raw material into the turntable, which is activated by the extension of the piston and latched until the button that is

below is pressed, which leaves the piece under the vertical piston.

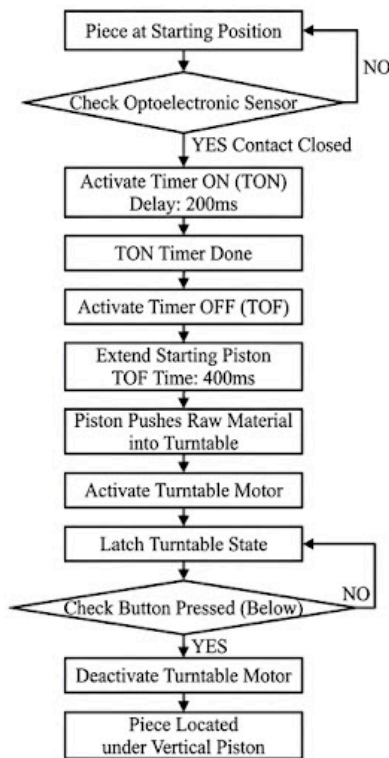


Figure 16. Flow Chart for Input Raw Material

Depending on the current piece that is being manufactured (chocolate, vanilla or strawberry) a multiplexor is used to set the corresponding number of presses the piston must perform; once it is done, the turntable is again activated and leaves the piece in position to be pushed to the first conveyor. When the piece pushes the conveyor button, the originally extended piston is contracted (with the use of a Pulse timer we ensure enough time is given for the piece to be pulled), placing the material on the conveyor and activating it.

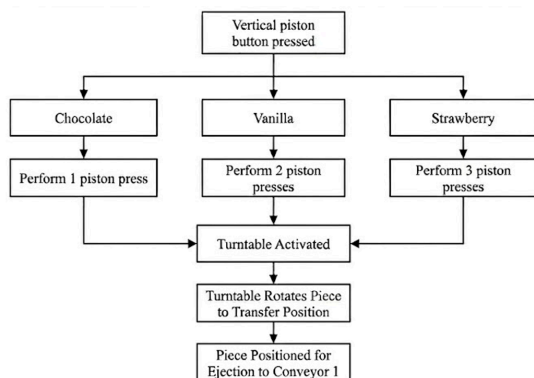


Figure 17. Flow Chart for Vertical Piston Operation

Since this point, all the manufacturing tasks, except for the cobot instructions are performed by the Manufacturing PLC, hence the state of the sensor placed at the end of the first conveyor and connected to the Pneumatic PLC has to be

sent via the S7 network. Following this, when the sensor detects the piece, the change produced at the contact is received by the Manufacturing PLC which activates the second conveyor and latches it. At this point the piece is transferred between both conveyor belts and then passes through the first optoelectronic sensor of the second conveyor, whose state is sent back to the Pneumatic PLC to indicate the first conveyor has already completed its task and must be turned off.

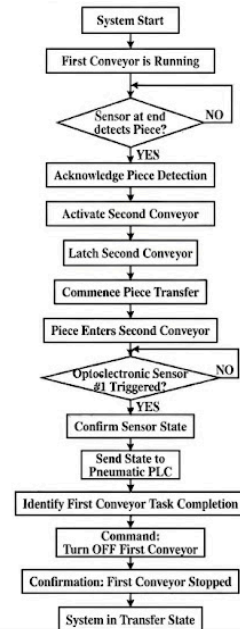


Figure 18. Flow Chart for Transfer Between Stations

When the piece reaches the last optoelectronic sensor of the second conveyor, two Pulse timers are chain triggered, one (3.5s) to keep the conveyor on for enough time to get the piece to the pusher position and the second one (3.3s) to delay the activation of the pusher that transfers the piece to the third conveyor. At the same time, the activation of the pusher is used to turn on the third conveyor which runs until its own optoelectronic sensor detects a piece, stopping it for the drilling operation.

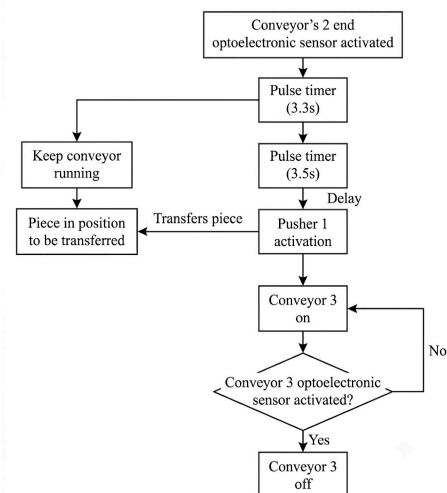


Figure 19. Flow Chart for Conveyor 2 to Conveyor 3 Transferring

An important feature of the programming is that for the machining part of the cell, data blocks with arrays going from 0 to 99 were created to store the type of the pieces detected with the vision sensor in the same order they were scanned, so we can keep track of which type of piece is been and will be manufactured at every time and the processing times are not mixed up.

This was achieved by creating inside the data blocks, two indexes, one that is used to keep track of the upcoming empty position (by augmenting in one whenever a position is used) of the array and store there the corresponding ID every time the vision sensor makes a classification, and the other one used to keep track of the piece that is currently being manufactured and set the corresponding times for each operation of the cell. It is also important to highlight that a data block with these arrays and indexes was created for each operation (drilling and mixing) and for the cobot to also keep track of the piece currently on the line.

With this information, the moment that the piece stops under the drill, a Pulse timer (whose preset time is selected by a multiplexor) is activated for the time corresponding to the piece type that is on the line; once the timer reaches its preset value, another Pulse timer is triggered for 2 seconds to move the conveyor again and transfer the piece to the mixing station conveyor (conveyor 4), where, following the same logic, the process is repeated; the conveyor carrying the piece stops when the corresponding optoelectronic sensor changes its state; with the ID stored at the array and the use of a multiplexor, the operation time is set on a Pulse timer, and once the operation is done, another Pulse timer activates the conveyor to move the piece to where the second pusher is located. Additionally the timer's output bit is used at the input of a timer ON that delays the activation of the pusher until the piece reaches the target position.

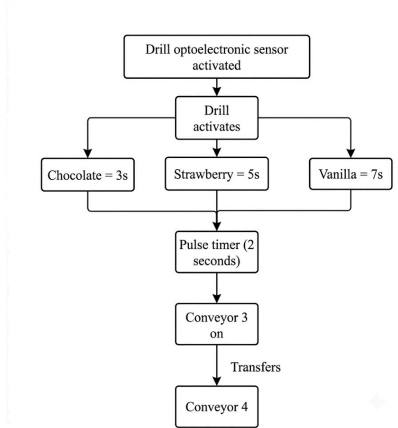


Figure 20. Flow Chart for Drilling Operation

When the pusher is activated, conveyor 5 is turned on to move the manufactured piece towards its edge, where an optoelectronic sensor stops it, sets a byte corresponding to decimal 4 and sends the first three bits to the cobot, indicating the piece has to be transported from conveyor 5 to conveyor 6 for the last part of the manufacturing process. The cobot grabs the piece from conveyor 5, releases it at the beginning of conveyor 6 and then retracts to a safe position,

when this routine is completed it sends three bits through its digital outputs to the PLC corresponding to a decimal 1, with the use of a comparator, a timer ON is triggered and after 300 milliseconds conveyor 6 is activated to move the piece towards the press machine.

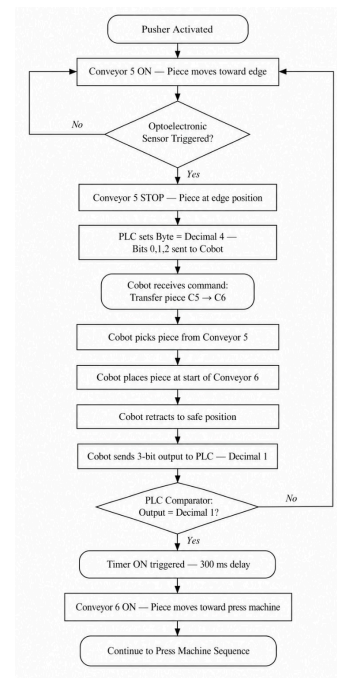


Figure 21. Flow Chart for Conveyor 5 to Conveyor 6 Transferring

With the use of a Pulse timer (800 ms) and a timer ON (300 ms), the piece is placed right below the press machine, and by using conveyor 6 end's optoelectronic sensor, the press machine is actuated down. After a period of 1.5 seconds set at a timer ON, the press machine is contracted and the piece is taken again to the start of the conveyor.

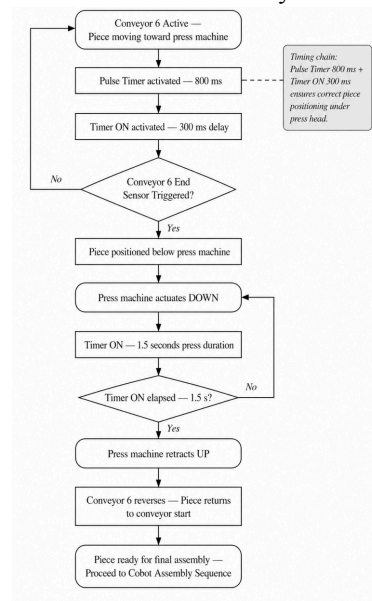


Figure 22. Flow Chart for Conveyor 6 and Press Machine Operation

By using the same logic as before, with a data block containing an array and a couple indexes, a multiplexor is used to set a byte corresponding to a decimal 5, 6 or 7 depending on the piece that is exiting the cell, then, the first three bits are sent through the digital inputs of the cobot so it grabs the matching complementary piece, assembles the final product and begins the palletizing sequence.

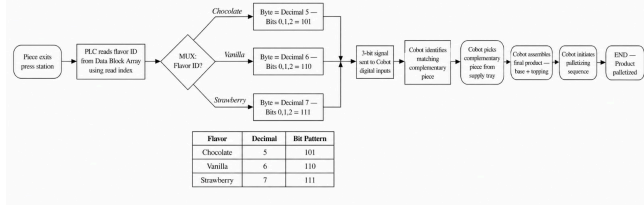


Figure 23. Flow Chart for Final Assembly and Palletizing

c. Collaborative Robot Control Logic and Integration

i. System Architecture and Hardware Configuration

The UFACTORY xArm Lite 6 collaborative robot was integrated into Cell 3105 as the primary material handling and palletizing subsystem. Physical communication with the cell's central Programmable Logic Controller (PLC) is established through a bidirectional digital I/O interface comprising four configurable inputs (CI0–CI3) and three configurable outputs (CO1–CO3). The hardware operates under active-low logic for inputs and active-high logic for outputs.

Signal	Designation	PLC Address	Function	Active State
Input	CI0	Q8.0	-	Normally HIGH
Input	CI1	Q8.1	Flavor selection bit 0	Active LOW
Input	CI2	Q8.2	Flavor selection bit 1	Active LOW
Input	CI3	Q9.4	Conveyor 5 end	Active LOW
Output	CO1	I0.1	Response bit 0	Active HIGH
Output	CO2	I0.2	Response bit 1	Active HIGH
Output	CO3	I0.3	Response bit 2	Active HIGH

Table 1. Digital I/O Configuration

ii. Control Interface Rationale

While the native UFACTORY Blockly interface is suitable for basic, linear pick-and-place routines, it is suboptimal for complex industrial logic. Blockly operates on a rigid, single-threaded execution model that forces the robot into a synchronous "wait-and-act" paradigm. This limitation

renders it incapable of managing concurrent workflows, tracking real-time production data, or handling multi-device asynchronous communication.

To overcome these limitations, the control logic was engineered using Python 3 and the native xArm-Python-SDK (v1.11.0). This implementation replaces traditional linear sequencing with a pseudo-concurrent pipeline architecture. By leveraging the Python runtime environment, the system executes non-blocking I/O polling, manages up to two active work-in-progress (WIP) units simultaneously, enforces independent safety timeouts, and facilitates advanced error logging.

Furthermore, critical safety telemetry is processed via independent background threads managed through asynchronous hardware callbacks (register_error_warn_changed_callback and register_state_changed_callback). This multi-threaded configuration isolates fault monitoring from the primary control loop, ensuring immediate actuator shutdown during collision events or emergency stops without introducing latency in the main execution cycle.

iii. Operational Sequence

The control software manages the production line via a non-blocking Finite State Machine (FSM). Rather than processing a single unit sequentially from start to finish, the FSM allows up to two units to co-exist in different stages of the production line. The robot executes a single evaluation per iteration, prioritizing tasks dynamically to maximize throughput. Each waiting state incorporates a 120-second timeout to prevent terminal blocking under fault conditions.

Phase 1 — Flavor Detection

The product flavor is identified by continuously polling the digital inputs of the cobot. A 50 ms software sampling window ensures signal stability and filters out electrical noise before the classification is committed.

Flavor	CI2	CI1	GPIO Pattern	Y Coordinate (mm)
Chocolate	HIGH	LOW	[1,0,1,1,1,1,1]	−200.0
Vanilla	LOW	HIGH	[1,1,0,1,1,1,1]	−250.0
Strawberry	LOW	LOW	[1,0,0,1,1,1,1]	−300.0

Table 2. Flavor Encoding

The X coordinate for tray access is shared across all flavors: X_1ST = [5.0, 55.0, 105.0] mm for base pieces and X_2ND = [155.0, 205.0, 255.0] mm for lids, indexed by a per-flavor counter that cycles through positions 0–2.

Phase 2 — Base Pick-and-Place

Upon flavor identification, the robot executes a pick-and-place operation transferring the cake base from the first section of the supply tray to the funnel that marks the

start of the production line. The trajectory consists of seven precise waypoints.

#	Waypoint	X	Y	Z	Rx	Ry	Rz
1	Home	200.0	-100.0	200.0	180.0	0.0	-45.0
2	Approach	target_x	target_y	50.0	180.0	0.0	-45.0
3	Pick	target_x	target_y	10.0	180.0	0.0	-45.0
4	Retract	target_x	target_y	50.0	180.0	0.0	-45.0
5	Transit	200.0	110.0	200.0	180.0	0.0	-45.0
6	Place approach	-37.8	243.4	193.3	180.0	0.0	-45.0
7	Place	-37.8	243.4	144.7	180.0	0.0	-45.0

Table 3. Phase 2 Trajectory

Waypoint 1 utilizes an asynchronous call (wait=False) to initiate physical hardware acceleration immediately while subsequent target positions are evaluated in the Python runtime.

Phase 3 — Cell Transfer Synchronization

Following base placement on the cell, the robot releases the “cake base” and becomes available to execute other pipeline tasks. Concurrently, the FSM monitors the unit by polling input CI3 at 50 ms intervals. CI3 transitioning to LOW signals that the part has reached the end-of-travel sensor on Conveyor 5 and is mechanically ready for inter-conveyor transfer.

Phase 4 — Inter-Conveyor Transfer

Upon CI3 activation, the cake base is transferred to Conveyor 6. The end-effector rotates from $R_z = -45^\circ$ (tray access orientation) to $R_z = 90^\circ$ (conveyor handling orientation) to align the cake base correctly for lid placement and prevent collisions. Upon placement at Conveyor 6 (252.1, -2.6, 56.4 mm), a 500 ms binary pulse is transmitted to the PLC via CO1–CO3 to signal that a base has been placed on the conveyor and to initiate the corresponding process.

Phase 5 — PLC Assembly-Ready Acknowledgment

The robot enters a non-blocking monitoring state, polling CII–CI3 for a flavor-specific acknowledgment handshake from the PLC. This handshake confirms that upstream processes are complete and the base is ready for final assembly.

Flavor	CI3	CI2	CI1	GPIO Pattern
Chocolate	LOW	HIGH	LOW	[1,0,1,0,1,1,1,1]
Vanilla	LOW	LOW	HIGH	[1,1,0,0,1,1,1,1]
Strawberry	LOW	LOW	LOW	[1,0,0,0,1,1,1,1]

Table 4. PLC Acknowledgment Encoding

Phase 6 — Topping Pick-and-Place

Once the PLC acknowledgment is validated, the topping corresponding to the active flavor is retrieved from the second section of the supply tray (X_2ND coordinates) and deposited on top of the cake base positioned at Conveyor 6 ($Z = 71.4$ mm). Upon successful placement, the specific per-flavor counter is incremented by one.

Phase 7 — Palletizing

The fully assembled product moves from Conveyor 6 onto a pallet, where it is stacked in a 2×2 grid across two vertical levels. To determine exactly where each product goes, the system uses a simple repeating sequence based on the total number of pallets processed:

$$grid_idx = global_pallet_count \pmod{4}$$

The height (Z-axis) automatically adjusts depending on the layer, shifting from Layer 1 ($Z = 10$ mm) to Layer 2 ($Z = 55$ mm). Once the product is placed, the system sends a 700 ms binary signal to log the specific product flavor into the PLC registry.

Flavor	CO3	CO2	CO1	Binary
Chocolate	0	1	0	010
Vanilla	0	1	1	011
Strawberry	1	0	0	100

Table 5. Palletizing Completion Codes

Phase 8 — Separator Lid Placement

To protect the products during storage, separator lids are inserted at two specific intervals. The system places an intermediate separator after every 4 units (picking at $Z = 80$ mm and placing at $Z = 40$ mm), and a final top lid after every 8 units (picking at $Z = 33$ mm and placing at $Z = 82$ mm). Once the top lid is in place, the global_pallet_count resets to zero to start the next palletizing cycle.

iv. Dynamic Task Scheduling and Pipeline Management

The system uses a priority-based scheduling method that allows the robotic cell to handle up to two products at the same time without needing complex multi-threading. Instead of processing one product from start to finish before starting the next, each product is tracked as an independent task within a Finite State Machine (FSM). Every task moves through its own series of steps, including base placement,

conveyor transfer, PLC synchronization, lid assembly, and palletizing.

To prevent traffic jams on the line and maximize efficiency, the scheduler checks all active tasks during every turn of the main control loop. Only one physical robot action happens per cycle, but the system constantly monitors every product using fast, non-blocking digital (GPIO) checks and timeout tracking. This allows multiple products to exist at different stages of the line while sharing the same robotic arm.

A Work-In-Process (WIP) limit of two active products is enforced through the parameter:

$$MAX_CONCURRENT = 2$$

This constraint was selected to prevent excessive queue buildup. As a result, one product may be waiting for conveyor transport or PLC acknowledgment while another product is already entering production. For example, if Task A is in State 3 (ready for lid assembly) while Task B is in State 4 (ready for palletizing), the scheduler will execute Task B's palletizing operation first (Priority 1), even if Task A has been waiting longer. This prevents assembly station blockage.

The scheduler applies a fixed-priority strategy designed to protect critical resources and avoid bottlenecks.

Priority	Operation	Objective
1	Palletizing completed products	Free the assembly station immediately
2	Conveyor 5 → Conveyor 6 transfer	Prevent conveyor blockage
3	Lid assembly	Complete partially processed products
4	Start new production cycle	Utilize remaining capacity

Table 6. Cobot Task Scheduling Priorities

This hierarchy ensures that products further down the line are finished before new work is introduced. For example, if a fully assembled product is waiting to be palletized at the same time a new product needs to be moved from the conveyor, the system will always choose to palletize first. Clearing the end of the line first keeps the production flow stable.

An additional safeguard prevents the system from accidentally creating duplicate tasks. Before accepting a new production order, the scheduler checks that another active task with the same flavor and tray position is not already running. This prevents repeated PLC signals from creating "ghost" products in the software and keeps the virtual tracking perfectly synced with the actual physical inventory.

The resulting setup provides pseudo-concurrent operation: while the physical robot can only move to do one thing at a time, multiple products can progress through the pipeline simultaneously. This significantly improves factory throughput compared to traditional systems, where a product has to exit the entire line before the next one can even begin.

v. Error Handling and Safety Mechanisms

To ensure industrial-grade reliability and protect the hardware, the system uses a four-tier safety framework:

- **State-of-Health Validation (is_alive):** The system continuously evaluates controller integrity by checking the physical connection status, confirming that `error_code == 0`, and verifying that the robot state remains operational (`state < 4`). It also includes a brief time filter to ignore temporary delays caused by motion buffering (State 5), which prevents unnecessary shutdowns.
- **Inline Return-Code Verification (_check_code):** Every motion and gripper command is wrapped in a validation function. If the xArm API returns any error code (a non-zero value), the system instantly sets the alive status flag to False, stops the robot's movement, and saves a detailed log report for troubleshooting.
- **Asynchronous Hardware Callbacks:** Critical events (such as emergency stops, safety curtain breaches, or motor over-current faults) trigger low-level background safety responses. Because these background threads run independently of the main Python program, they can immediately turn off the operational flag and lock the motor brakes the moment a safety limit is crossed.
- **Monotonic Timeouts:** To prevent the robot from freezing due to a broken sensor or a missing PLC signal, all waiting periods are managed by strict software deadlines using `time.monotonic()`. If the system gets stuck waiting for more than 120 seconds, it automatically runs an orderly shutdown sequence: it turns off all digital outputs and brings the robotic arm to a safe stop so an operator can step in.

vi. Pseudocode and State Transitions

To clarify how the priority strategy works in practice, the core logic of the main control loop can be represented by the following structured pseudocode:

```

INITIALIZE cobot
INITIALIZE task list

WHILE system is operational:

    # Clean up the pipeline
    Remove completed tasks from list

    # Check background PLC status for waiting tasks
    FOR each task waiting for PLC acknowledgment:
        IF acknowledgment received:

```

```

        Move task to "Ready for Lid Assembly"
    ELSE IF timeout expired:
        Trigger safety stop
    END IF
END FOR

# -----
# Priority 1: Palletize completed products
# -----
IF any task is ready for palletizing:
    Palletize product
    Place separator lid if counter threshold is met
    Mark task as completed

# -----
# Priority 2: Transfer product between conveyors
# -----
ELSE IF any task is waiting for Conveyor Sensor CI3:
    IF CI3 activated:
        Transfer product from Conveyor 5 to Conveyor 6
        Notify PLC of transfer
        Set task state to wait for PLC acknowledgment
    ELSE IF timeout expired:
        Trigger safety stop
    END IF

# -----
# Priority 3: Assemble topping
# -----
ELSE IF any task is ready for lid assembly:
    Pick topping/lid from storage
    Assemble topping onto product base
    Mark task as "Ready for Palletizing"

# -----
# Priority 4: Start new product
# -----
ELSE IF active products < MAX_CONCURRENT:
    IF new flavor detected by entry sensors:
        Pick new cake base
        Deposit base onto Cell
        Register new task in FSM tracking list
    END IF
END IF

# Prevent CPU saturation during idle cycles
IF no physical action was executed during this cycle:
    Sleep briefly

END WHILE

```

Each task registered in the scheduler moves through a strict, sequential Finite State Machine (FSM). The diagram below illustrates the life cycle of a single product as it progresses through the cell:

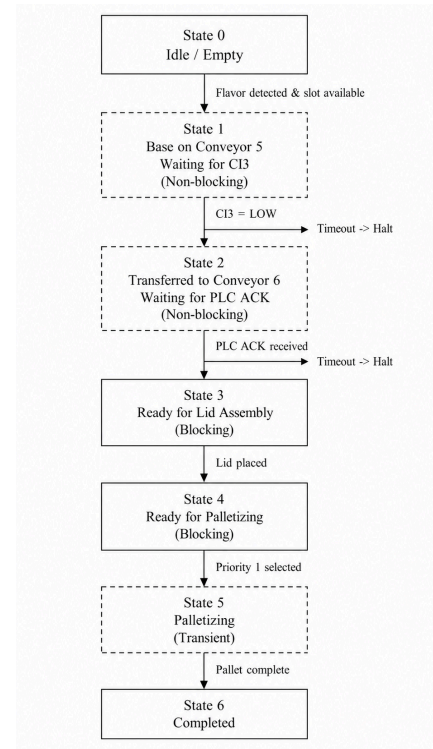


Figure 24. Finite State Machine (FSM) Diagram of the Single-Product Lifecycle

C. Cell HMI Interfaces

The HMI system consists of two independent panels: the Pneumatic HMI and the Machining HMI. Both share a consistent visual language and a unified navigation flow, operating as a coordinated interface for the production cell.

At startup, both screens display a locked access screen. The operator must present a valid RFID card to the reader to authenticate. Once the credentials are verified, the Pneumatic HMI displays the recognized user's name and assigned role, allowing the operator to confirm their identity before pressing Enter to gain access. Upon doing so, both HMIs transition simultaneously to their respective main screens, ensuring session state remains synchronized across both panels.

The main screen serves as the central navigation hub. On the Pneumatic HMI, it provides access to five screens: Plant View, Set Process Times, Manual Control, Camera, and Get Cycle Time. The Machining HMI main screen gives access to three screens: Plant View, Set Process Times, and Manual Control. Each navigation button includes a status lamp that indicates access permission: green signals that the current user is authorized to enter that screen, while grey indicates restricted access. A button with no permission produces no response when pressed, preventing unauthorized navigation without displaying error prompts that could distract the operator.

To return to the main screen from any sub-screen, the operator presses the Home button, which is consistently placed across all screens. Session termination is handled through the Logout button located at the bottom left of the

Pneumatic HMI's main screen; pressing it ends the active session on both panels simultaneously and returns them to the access screen, ready for the next user.

The interface design follows a minimalist approach, using grey as the dominant background color across all screens. Color is used purposefully and sparingly: red is reserved exclusively for warning alarms, and green for operational status indicators. This controlled use of color reduces visual noise and ensures that critical signals are immediately distinguishable from the rest of the interface. All screens use Tahoma as the sole typeface, reinforcing visual consistency. Text is sized to remain legible at operator viewing distance, and buttons are dimensioned to allow reliable interaction on a touchscreen panel, minimizing the risk of accidental input.

The interface design complies with the ANSI/ISA-101 Human-Machine Interface standard for industrial applications. The use of a neutral grey background with color reserved only for alarms and status indicators aligns with the standard's guidelines on color philosophy, which discourage decorative use of color in order to preserve its communicative value. The consistent navigation structure, a central main screen with clearly labeled sub-screens and a fixed Home button, reflects the standard's emphasis on predictable, low-cognitive-load navigation. Role-based access control with visual feedback through permission lamps supports the standard's requirements for operator guidance and error prevention. The uniform font, button sizing, and layout hierarchy across both panels further satisfy ISA-101's principles of interface consistency and ergonomic design.

a. Pneumatic PLC

Access: The screen that shows on startup. It waits for the user to place an authorized RFID card on the reader, once read, current user and role are displayed and the user is free to press the enter button and gain access to the main screen of both the pneumatic and machining PLC.

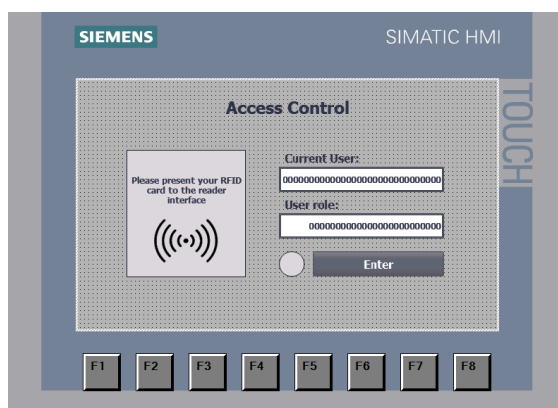


Figure 25. Access Control Screen of the Pneumatic Station HMI

Main Screen: Once authenticated, the operator is taken to the main screen, which serves as the central navigation hub. It displays the complete list of available screens as labelled buttons, each accompanied by a square indicator lamp. If the

operator has access to a given screen, the lamp turns green; if access is restricted, it remains grey and the button produces no response when pressed, silently preventing unauthorized navigation. A cake graphic is displayed on the right side of the screen to represent the company's line of business. The Logout button is located at the bottom left, ending the active session when pressed.

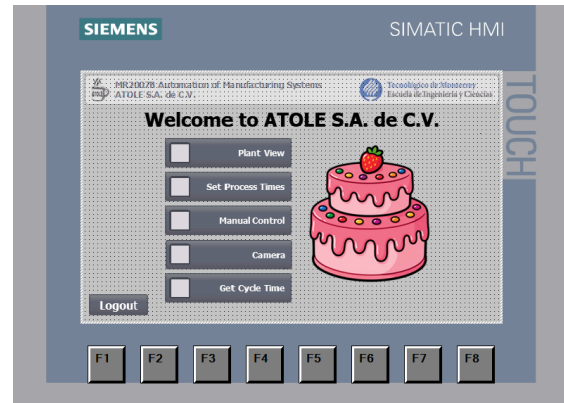


Figure 26. Main Navigation Screen of the Pneumatic Station HMI

Manual Control: The Manual Control screen allows individual actuation of the pneumatic components within the plant. Manual mode can be enabled using the switch located on the left side of the screen. Once enabled, the screen displays a list of buttons corresponding to each pneumatic actuator available for manual operation. Pressing a button energizes the associated actuator, while releasing it causes the actuator to return to its inactive state. An indicator lamp is displayed next to each actuator to show its current status, allowing the operator to verify whether the corresponding actuator is activated or deactivated.

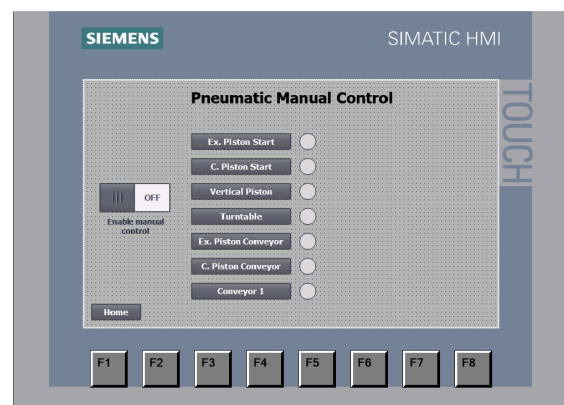


Figure 27. Manual Control Interface for Pneumatic Actuator Operation

Camera Control: The Camera Control screen is used to initiate the manufacturing process. When the operator presses the Scan Workpiece button, a signal is sent to the Cognex camera to capture an image of the work area. Once the camera data is received, the screen displays the ID of the scanned workpiece, the cake flavor associated with that ID, and the elapsed time, in seconds, since the image was

captured. If a valid ID is detected, a signal is sent to the cobot, which proceeds to pick up the workpiece corresponding to the identified flavor. An alarm message is displayed above the scan button whenever the cobot or the pneumatic station is busy, preventing the operator from initiating a new scan while the system is occupied. While this alarm is active, the Scan Workpiece button is disabled and pressing it has no effect.

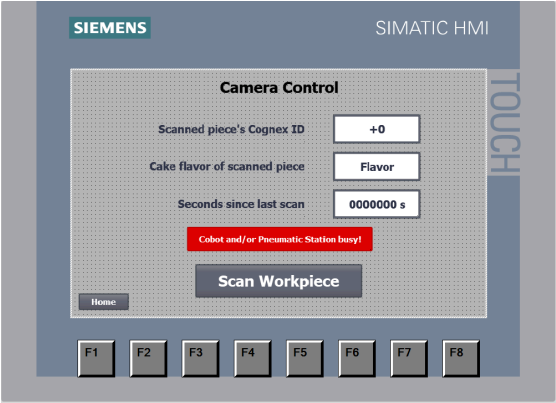


Figure 28. Camera Control Screen for Workpiece Identification and Production Initiation

Plant View: The Plant View screen provides a top-down representation of the pneumatic station. The color scheme was intentionally designed to resemble the physical plant as closely as possible, allowing operators to easily associate the graphical elements displayed on the HMI with their real-world counterparts. This approach enhances visual recognition of the equipment and reinforces the relationship between the physical system and its digital representation. All actuators are displayed in their corresponding locations within the station, each accompanied by indicator lamps that show their current status. For actuators controlled by double-solenoid valves, two indicator lamps are provided to represent the extension and retraction solenoids independently. The live pressure of the pneumatic system is displayed in pounds per square inch (PSI). Additionally, a power consumption indicator located in the lower-right corner of the screen displays the instantaneous power consumption of the pneumatic station and cobot in watts.

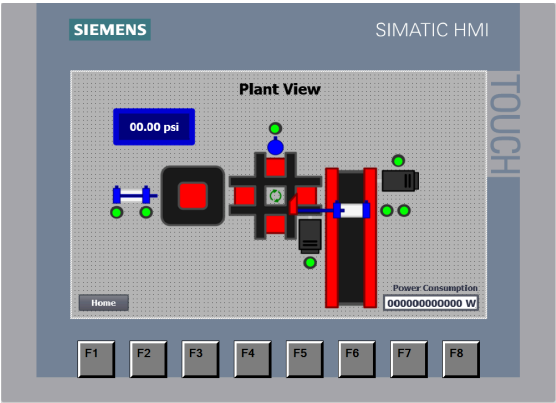


Figure 29. Plant View Screen Showing the Pneumatic Station Layout and Real-Time Status Indicators

Process Times: The Process Times screen allows the operator to configure the pressing duration of the vertical pneumatic piston for each cake flavor. Separate time values can be defined for chocolate, vanilla, and strawberry products, enabling the pressing operation to be adjusted according to the specific requirements of each flavor. These values are used by the control system during automatic operation to determine how long the vertical piston remains actuated while processing each workpiece.

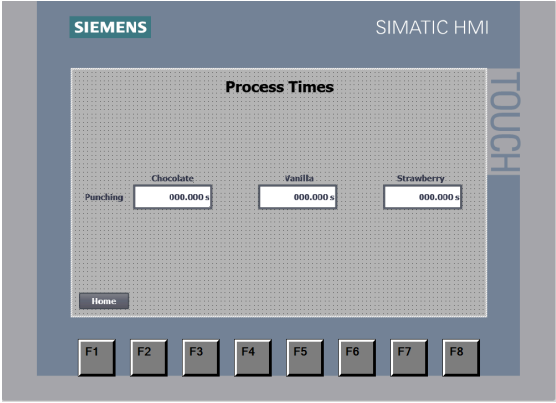


Figure 30. Process Time Configuration Screen for Flavor-Dependent Pressing Operations

Measure Cycle Time: The Measure Cycle Time screen allows the operator to initiate a cycle time measurement routine. Measurement mode is enabled by pressing the Enable Measuring button, causing the associated indicator lamp to turn green and signal that the system is ready to record cycle time data. Once measurement mode is active, the operator can scan a workpiece and start a manufacturing sequence. During operation, the screen displays the elapsed time of the current cycle as well as the ID of the workpiece being processed. A configurable target cycle time is provided at the bottom of the screen, allowing the operator to define a reference value for performance evaluation. Instructions describing the proper use of the measurement routine are displayed on the right side of the screen.

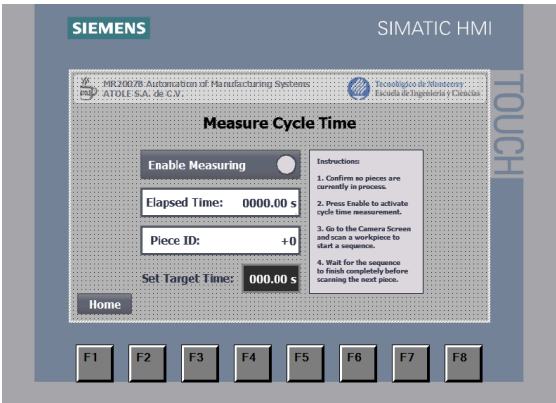


Figure 31. Cycle Time Measurement Interface for Production Performance Evaluation

b. Machining PLC

Access: The Access Control screen does not provide any operational controls, indicators, or configurable parameters. Its sole purpose is to prevent unauthorized users from accessing the HMI. Authentication cannot be performed from this screen; instead, user authentication is carried out through the Access Control interface of the Pneumatic HMI. Until a valid authentication is completed on that system, access to the remaining screens and functions of this HMI remains restricted.

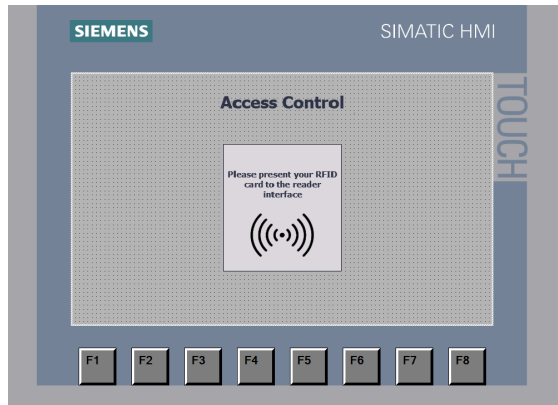


Figure 32. Access Control Screen of the Machining Station HMI

Main: Following successful authentication through the Pneumatic HMI, the operator is directed to the Main screen of the machining station, which serves as the central navigation hub for this HMI. The screen displays the complete list of available screens as labeled buttons, each accompanied by a square indicator lamp that reflects the access permissions assigned to the current user. If access to a given screen is granted, the corresponding indicator turns green; otherwise, it remains grey and the associated button produces no response when pressed. A cake graphic is displayed on the right side of the screen to represent the company's line of business. Unlike the corresponding screen in the Pneumatic HMI, no Logout button is provided, as user authentication and session management are handled exclusively through the Pneumatic HMI.

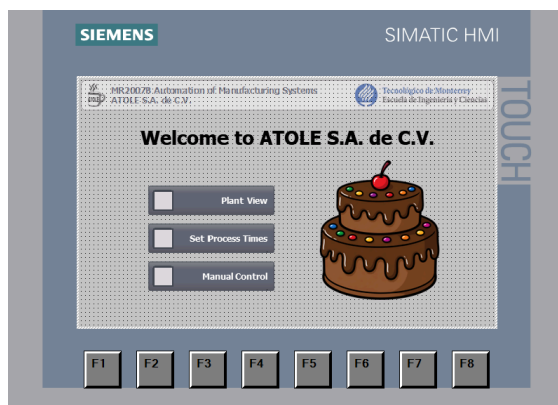


Figure 33. Main Navigation Screen of the Machining Station HMI

Manual Control: The Manual Control screen provides the same functionality as the corresponding screen in the Pneumatic HMI, but for the actuators located in the machining station. Manual mode can be enabled using the switch located on the left side of the screen. Once enabled, the operator may actuate any available component by pressing its corresponding button. An indicator lamp is displayed next to each actuator to show its current status, allowing the operator to verify whether the corresponding actuator is activated or deactivated.

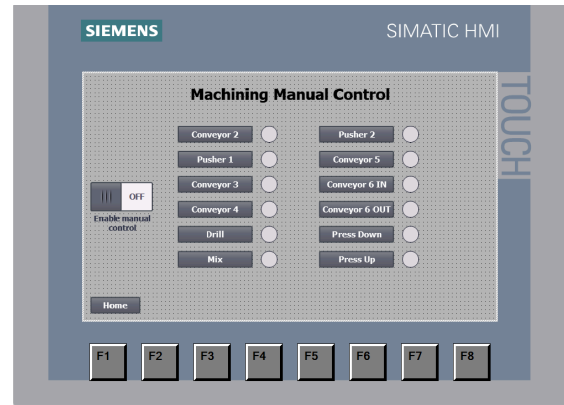


Figure 34. Manual Control Interface for Machining Station Actuators

Plant View: The Plant View screen provides the same functionality as the corresponding screen in the Pneumatic HMI, presenting a top-down representation of the machining station and its actuators. Each actuator is displayed in its corresponding location within the station and is accompanied by an indicator that shows its current status. For actuators capable of bidirectional operation, such as Conveyor 6 and the motor driving the stamping mechanism, directional arrows are used instead of a single status indicator to represent the current direction of motion. Additionally, a power consumption indicator located in the lower-right corner of the screen displays the instantaneous power consumption of the machining station in watts.

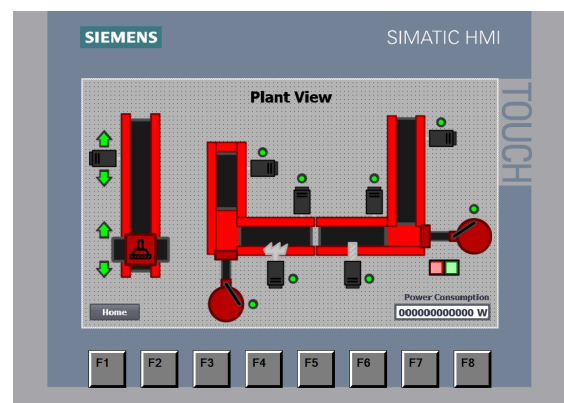


Figure 35. Plant View Screen Showing the Machining Station Layout and Real-Time Status Indicators

Process Times: The Process Times screen provides the same functionality as the corresponding screen in the Pneumatic HMI, allowing the operator to configure process-specific timing parameters for each cake flavor. Separate time values can be defined for the drilling and mixing operations of chocolate, vanilla, and strawberry products. These values are entered in seconds and are used by the control system during automatic operation to determine how long each workpiece remains at the corresponding machining station.

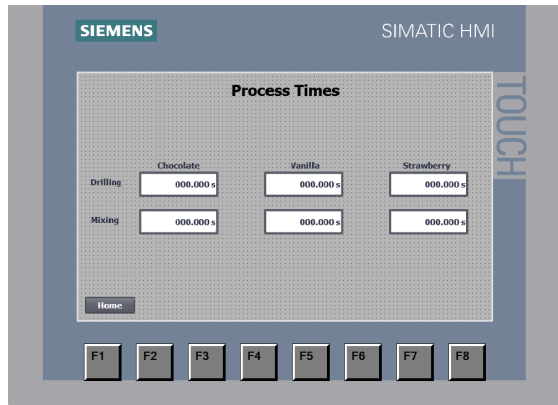


Figure 36. Process Time Configuration Screen for Drilling and Mixing Operations

D. Monitoring Centers (Dashboards) & Simulation

a. Plant Dashboards

To provide centralized monitoring of both manufacturing environments, three independent dashboards were developed in Node-RED. The dashboards were integrated into a single web-based platform accessible through different tabs, enabling operators and supervisors to navigate between Plant 1 (CIMA), Plant 2 (Cell 3105), and a global monitoring interface.

The dashboards were designed following an Industrial Internet of Things (IIoT) architecture in which data from the ESP32 gateway and the PLC-based automation cell were collected, processed, and transmitted through MQTT communication. The ROMI T240 monitoring system published data directly from the ESP32 to the MQTT broker, while the manufacturing cell data was acquired from the PLC through a Python-based interface that acted as an intermediary between the PLC and the MQTT infrastructure. This resulted in two independent data acquisition systems converging into a unified monitoring platform.

i. Plant 1 Dashboard – CIMA IoT Monitoring

The Plant 1 dashboard was specifically designed to monitor the electrical behavior and energy performance of the ROMI T240 lathe during machining operations. The interface served as the primary monitoring tool for the IoT instrumentation system, providing operators with both real-time measurements and historical performance data.

The dashboard was organized into three main sections. The first section displayed key performance indicators (KPIs),

including current, power consumption, energy consumption, and estimated operating cost. These indicators were updated continuously from the measurements acquired by the ESP32 gateway and provided an immediate overview of the machine's electrical demand.

The second section consisted of historical trend charts showing current, power, energy, and cost as functions of time. These visualizations enabled the identification of machining cycles, transient events, startup conditions, and variations in energy consumption. By analyzing these trends, users could correlate electrical demand with specific manufacturing activities and evaluate the efficiency of the machining process.

The third section focused on machine operational status. The dashboard displayed the current machine state (OFF, IDLE, CUTTING, or FAULT) together with accumulated operating times for each state. This information allowed users to evaluate machine utilization, identify productive and non-productive periods, and estimate equipment availability throughout the manufacturing cycle.

In addition to real-time monitoring, all process variables were archived in CSV files and SQL databases, enabling subsequent analysis of machine utilization, energy efficiency, and production performance.



Figure 37. Plant 1 Dashboard for Real-Time Monitoring of the ROMI T240 Lathe

ii. Plant 2 Dashboard – Cell 3105 IoT Monitoring

The Plant 2 dashboard was developed to provide comprehensive supervision of the automated manufacturing cell. Unlike the Plant 1 dashboard, which focused on energy monitoring, this interface concentrated on operational status, production performance, and subsystem coordination.

The dashboard consolidated information from multiple cell components, including the pneumatic station, machining station, collaborative robot, and production management system. Data was acquired from the PLC and transmitted through the MQTT infrastructure using a Python-based communication bridge.

The interface was divided into functional areas that reflected the physical organization of the cell. A dedicated section monitored the machining and pneumatic stations by displaying their instantaneous power consumption and pneumatic pressure values.

Safety-related information was also incorporated into the dashboard through emergency status indicators associated with both the machining and pneumatic stations.

Production monitoring was implemented through counters that tracked the number of completed parts delivered for each product category (Chocolate, Vanilla, and Strawberry) as well as the total production output. These counters provided real-time visibility of manufacturing progress and facilitated production tracking during laboratory operation.

A dedicated area displayed the status of the collaborative robot, indicating whether the cobot was actively executing a motion sequence or waiting for the next task. User-management information was also integrated into the dashboard, displaying the currently logged-in operator and associated user role. This functionality supported traceability and improved accountability during cell operation.

Furthermore, operational indicators were included through the visualization of total energy consumption and estimated operating cost for the automated cell. These metrics complemented the production indicators and enabled evaluation of both productivity and resource utilization.

Overall, the Plant 2 dashboard provided operators with a centralized interface for supervising equipment status, production throughput, utility consumption, safety conditions, and user activity within the manufacturing cell.

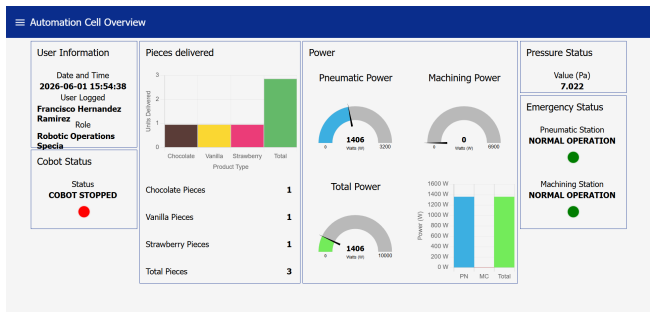


Figure 38. Plant 2 Dashboard for Monitoring and Supervision of Cell 3105 Operations

ii. Integrated Overview Dashboard

To facilitate system-wide supervision, an additional dashboard was developed to aggregate the most relevant information from both manufacturing environments into a single interface. This overview dashboard acted as a monitoring center, providing a consolidated representation of the entire manufacturing system.

The dashboard displayed key operating metrics from the ROMI T240 lathe, including machine state, current consumption, power demand, energy usage, and operating cost. Simultaneously, it presented the operational status of Cell 3105, including cobot activity, machining and pneumatic station conditions, production counts, and user information.

By integrating information from both plants, the overview dashboard allowed users to evaluate the overall condition of the manufacturing system without navigating between multiple interfaces. The centralized architecture also demonstrated the interoperability of the proposed IIoT framework, successfully integrating embedded IoT devices, PLC-based automation systems, MQTT communication, databases, and web-based visualization tools into a unified monitoring environment.

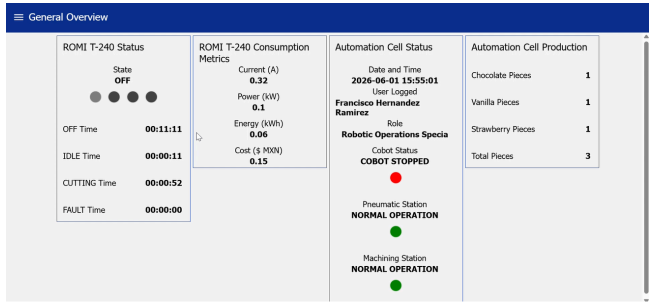


Figure 39. Integrated Overview Dashboard for Unified Monitoring of the Manufacturing System

The developed dashboards can be accessed through the following URL: <http://10.25.110.236:1880/ui>, provided that the workstation hosting the Mosquitto MQTT broker is active and connected to the institutional network. This capability allows authorized users to monitor the manufacturing systems from different locations while maintaining real-time access to operational and energy-related data.

Throughout the project, the dashboards underwent multiple design iterations. In addition to the initial implementation, a second version was developed with the objective of improving usability, enhancing information organization, and facilitating user interaction. The updated interfaces incorporated additional process indicators, improved visual layouts, and more comprehensive operational metrics, allowing operators to access relevant information more efficiently. These modifications contributed to a more intuitive monitoring experience and provided a broader understanding of both manufacturing processes. The final versions of the dashboards are presented in the following figures.

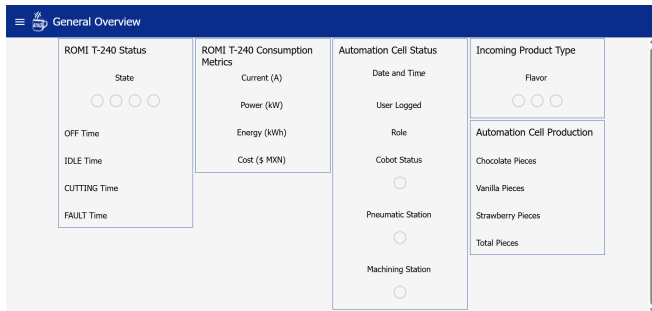


Figure 40. Integrated Overview Dashboard V2

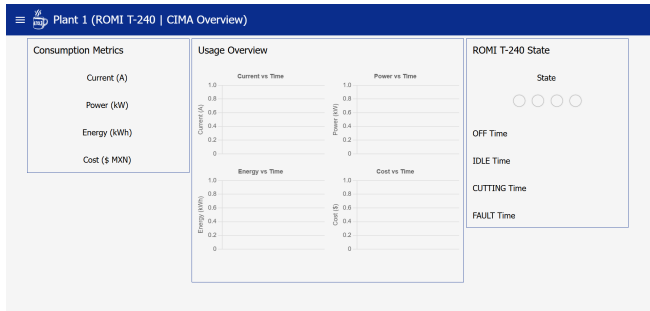


Figure 41. Plant 1 Dashboard V2

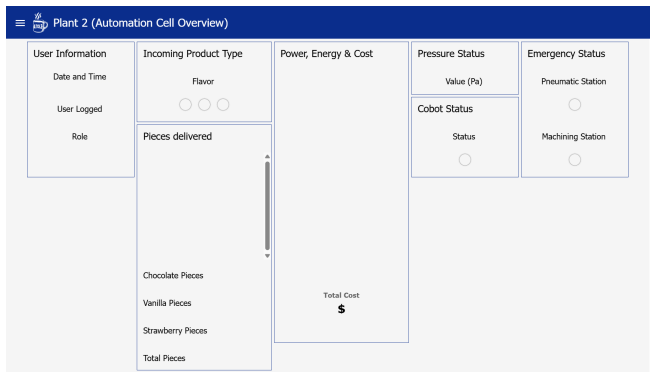


Figure 42. Plant 2 Dashboard V2

b. Process Simulation

i. Description of the complete production line

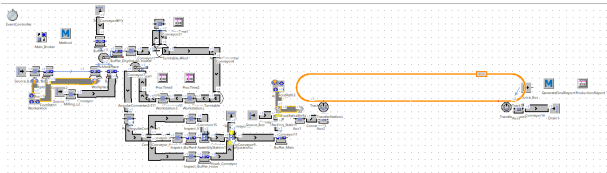


Figure 43. Complete 2D Layout of the Production Plant Simulation

The production line simulation was developed in Tecnomatix Plant Simulation to accurately replicate the physical manufacturing process. The simulation begins with Line 1, where cylindrical parts (blue material) are manufactured using a lathe machine operated by Worker 1. The lathe station (Lathe_L1) was configured with a processing time of 1 hour, 47 minutes, and 57 seconds per batch, with automatic processing enabled to simulate continuous operation. Material supply for this line was modeled using Source_L1, which generates batches of 10 cylinder units (MU_Cylinder).

Line 2 operates simultaneously using a fully automatic milling machine (Milling_L2) to produce V-shaped parts (orange material). In accordance with the challenge specifications, since Line 1 utilizes a lathe, Line 2 employs a milling machine with a processing time calculated as half of the lathe's cycle time, resulting in 53 minutes and 58.5 seconds per batch. The milling station was configured with

an 8-minute setup time, a Mean Time To Repair (MTTR) of 10 minutes, and 85% availability to reflect realistic operational conditions. Source_L2 was programmed to supply batches of 20 V-shaped parts (MU_VShape) to this line.

The machined cylinders from Line 1 are initially stored in an initial buffer (Buffer), which was configured with a capacity of 20 units using a queue-type buffer structure. Worker 2 was then assigned to transfer these parts from Buffer to Buffer1, maintaining the same storage capacity of 20 pieces. Simultaneously, the V-shaped parts from Line 2's milling machine are transported via conveyor belt to Buffer2, which also maintains a capacity of 20 units.

Additionally, a third production stream was incorporated into the simulation to introduce hook components (MU_Hook) into the manufacturing process. This supplementary line utilizes Source_Strawberry, configured to generate single units at intervals of 1 hour and 50 minutes, with production starting at 1:50:00 and stopping at the end of the simulation day (1:00:00:00). These hook parts are stored in Buffer3, which was configured with a capacity of 10 units in a queue-type structure. This third buffer enables the integration of an additional component into the assembly process, expanding the complexity of the material handling system.

The material selection process from these buffers using the first Pick and Place robot was implemented using custom CONTROLS and METHOD programming. The selection logic was coded to prioritize buffer retrieval based on availability: the system first checks if Buffer1 contains parts (Buffer1.numMU > 0), then Buffer2, and finally Buffer3, ensuring efficient material flow without bottlenecks.

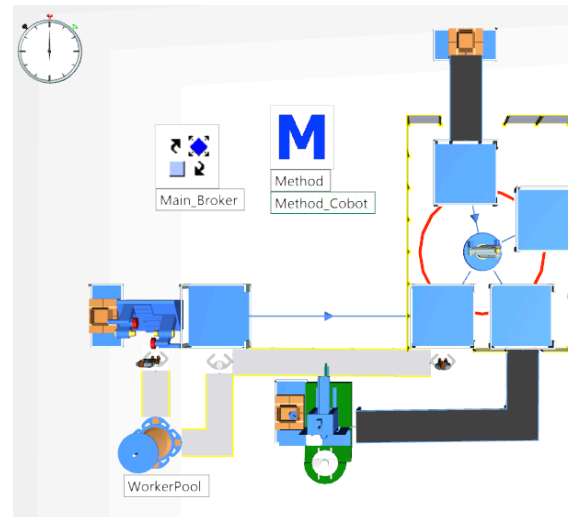


Figure 44. 3D Layout of the CIMA Production Line Simulation

The PickAndPlace COBOT is configured with loading and unloading times of 10 seconds each, set with a capacity of 1 unit, and configured to retain the MU (Material Unit) conveying direction. Once the COBOT retrieves a part, it is placed into Buffer_Entry, a queue-type buffer configured

with a capacity of 1 unit, which acts as the direct entry point to the manufacturing cell. From Buffer_Entry, the parts are immediately fed into the cell by a Hydraulic_Pusher station. The Hydraulic_Pusher is configured with automatic processing and a rapid processing time of 2 seconds.

After being pushed by the Hydraulic_Pusher into Conveyor1, the parts enter the manufacturing cell's conveyor system, which directs them toward the first processing stage (piston station). To accurately simulate the physical layout and belt direction changes required by the challenge, the material flow incorporates turntables and angle converters. The parts are first directed through a turntable (Turntable_4Pos), configured with Entry and Exit Angle Tables to simulate the 360° rotating turntable in the physical cell. In this case, the entry angle is 180° and the exit angle is 270° into Conveyor2.

Following this directional change, the parts are transported along the conveyor to Workstation1. This workstation is set to automatic processing, with its specific processing times dynamically controlled via a List(Type) variable named "ProcTime1". This table assigns distinct processing times based on the part type: 2 seconds for MU_Cylinder, 6 seconds for MU_VShape, and 4 seconds for MU_Hook, accurately reflecting the varied treatment times defined for the manufacturing cell operations.

Upon exiting Workstation1, the parts are directed into a second turntable (Turntable_4Pos1). This second turntable is precisely configured to redirect the flow, ensuring the parts are correctly oriented and physically redirected onto the subsequent conveyor belt, Conveyor3. To fulfill the specific simulation requirement of using at least one angle converter alongside the turntables, the parts on Conveyor3 are then routed through an Angular Converter. This component facilitates the transition of the parts onto Conveyor4, adjusting their orientation for the next stage of the cell.

Conveyor4 then directs the parts into a final turntable (Turntable), which is configured with specific Entry and Exit Angle Tables to manage the final directional shift. The Entry Angle Table is set to receive parts from Conveyor4 at an angle of 90 degrees, while the Exit Angle Table directs them onto Conveyor5 at an angle of 0 degrees. Finally, the parts are transported along Conveyor5 to reach Workstation2, the second stage of the manufacturing cell.

Following transport via Conveyor5, the parts arrive at Workstation2, labeled as the Drilling station. This station is configured for automatic processing with no setup or recovery times. The processing duration is dynamically controlled via a List(Type) variable named ProcTime2. This table assigns distinct processing times based on the specific part type entering the station: 3 seconds for MU_Cylinder, 7 seconds for MU_VShape, and 5 seconds for MU_Hook.

Upon completion of the drilling operation, the parts are transferred to the final stage of the manufacturing cell, Workstation3, labeled as the Mixing station. This workstation is also set to automatic processing. Its cycle

times are set by the ProcTime3 table, which similarly differentiates processing durations for each component (3 seconds for cylinders, 7 seconds for V-shapes, and 5 seconds for hooks), completing the treatment phase before the parts exit the manufacturing cell to be picked up by the second COBOT.

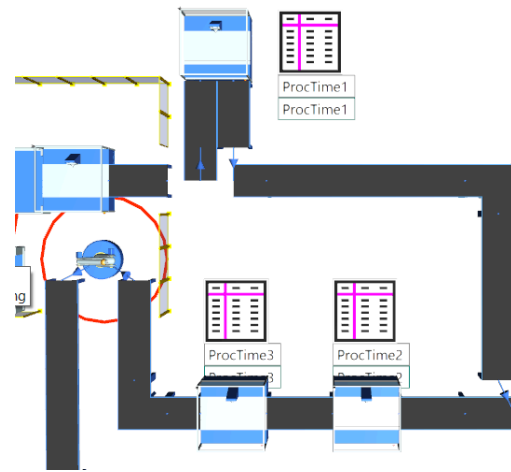


Figure 45. 3D Layout of the Manufacturing Cell

The parts are then placed onto the polishing belt and fed into the Polishing station. This workstation is configured for fully automatic processing with a constant cycle time of 10 seconds per part. No setup or recovery times are applied at this stage, ensuring a continuous surface-finishing flow.

Once the polishing process is complete, the parts are transported along the conveyor to a Converter designed to physically separate the different component types. This converter is configured with the "MU Name" strategy, utilizing a custom AttributeList to route each specific part to its designated exit conveyor: MU_Hook is directed to exit 1, MU_VShape to exit 3, and MU_Cylinder to exit 0.

Following this separation, each part type is routed to its dedicated inspection station (Inspect_Cylinder, Inspect_V, and Inspect_Hook). All three inspection stations are configured for fully automatic processing with a constant cycle time of 30 seconds per part, ensuring that every component undergoes a standardized quality control check before proceeding.

The validated cylinders are routed to intermediate storage buffers to synchronize the differing production rates and prevent line overflow. Specifically, they are routed to Buffer4, a queue-type buffer configured with a capacity of 15 units. From this buffer, the cylinders proceed to AssemblyStation1. This station is configured to perform the initial sub-assembly, combining 1 Cylinder (acting as the Main MU from Predecessor 1) with 2 V-shapes (Predecessor 2). The station is operated in "Delete MUs" mode, consuming the individual parts to create the sub-assembly.

Simultaneously, the inspected hooks are stored in a parallel buffer (also with a capacity of 15) to ensure availability for the final assembly stage. Additionally, a separate material

stream is introduced via Source_CakeTop, which generates MU_CakeTop units at a fixed interval of 2 hours (02:00:00).

At AssemblyStation2, these three streams (the sub-assembly from Station 1, the hooks, and the cake tops) are combined. This final assembly station is configured with the sub-assembly as the Main MU (Predecessor 1). The assembly table dictates the addition of 1 Hook (from Predecessor 2) and 1 CakeTop (from Predecessor 3). The completed products are stored in a buffer configured with a capacity of 20 pieces. This buffer serves to decouple the assembly line from the packing process, ensuring a steady supply of products even if packing is momentarily delayed.

The packing operation is managed at the Packing_Station. To support this, a separate material stream is introduced via Source_Box, which supplies packaging units (.UserObjects.Box) at a fixed interval of 2 hours and 10 minutes. These boxes are transported via conveyor to the packing station. At the Packing_Station, the system is configured to combine the final assembly (acting as the Main MU from Predecessor 1) with the packaging box (from Predecessor 2). The station utilizes the "Attach MUs" assembly mode, which physically groups the assembly and the box into a single transport unit for the final stage.

This operation is performed by Worker_3, who is assigned to the station's services list. The sequence is set to "MUs then Services," ensuring that both the assembly and the box are present before the worker begins the task. The packing process is set to take 10 seconds per unit.

Once packed, the boxed assemblies are transferred from the Packing_Station to auxiliary processing stations before final distribution. The packed units first enter Aux1, an auxiliary station configured with automatic processing and a cycle time of 10 seconds. This station is linked to Workplace4, where workers can perform additional tasks if required, though the "Worker stays here after completing the job" option is disabled to allow worker mobility.

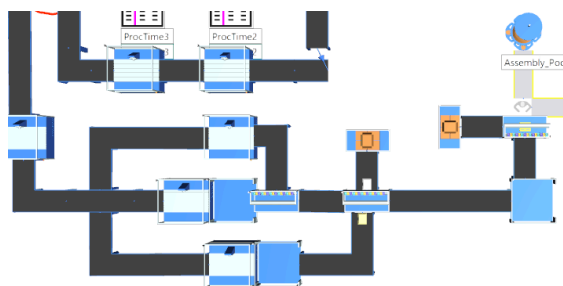


Figure 46. 3D Layout of the Inspection and Assembly, and Packing Areas

Following Aux1, the boxes are transferred to Aux2, another auxiliary processing station also configured with a 10-second automatic processing time. These auxiliary stations are utilized to ensure proper spacing and sequencing of packed units.

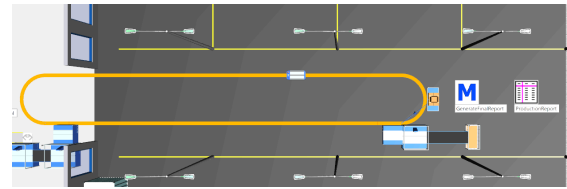


Figure 47. 3D Layout of the Shipping and Distribution Segment

From Aux2, the packed boxes are transferred to the plant's main transport network via TransferStationL. This transfer station is configured as a "Load" type station, which physically loads the parts from Aux2 onto the track system (bus network). The sensor position is set to Lane A at position 2, ensuring precise detection and routing of the packed assemblies.

Once loaded, the packed boxes travel along a 30-meter Track. This track is configured with a length of 30 meters, a width of 0.3 meters, and an unlimited capacity (-1) to accommodate the continuous flow of packed units. To simulate realistic material handling along this route, a bus system is implemented using Source_Bus, which generates a single Bus unit (.UserObjects.Bus) at the 2:26:00 mark to transport the loaded packages along the track.

At the end of the transport route, the packages are unloaded from the track using TransferStationU. This station is configured as an "Unload" type, detecting the parts at sensor position 28 on Lane A and transferring them to the final auxiliary station, Aux3. Aux3 is configured with automatic processing and a cycle time of 10 seconds, serving as the final handling point before the products exit the system. Finally, the packed assemblies are moved to a Drain (Sink) object, marking the successful completion of the entire production and shipping simulation.



Figure 48. Complete 3D Visualization of the Production Plant

ii. Results Analysis

The simulation results demonstrate the successful completion of the production cycle over a simulated period of 1 day (1:00:00:00.0000). These statistics obtained via the Summary Report and a custom method (GenerateFinalReport) reveal critical insights into the system's performance and material flow efficiency:

- **Production Output Analysis:** The final statistics from Drain1 show that 10 packed boxes successfully completed the entire production process. The mean life time of these units was approximately 3 hours and 45 minutes, indicating

the total time from raw material input to final shipment. The production breakdown shows:

- 10 cylinders were machined on Line 1 (Lathe_L1)
- 20 V-shapes were machined on Line 2 (Milling_L2)
- 11 hooks were supplied via Source_Strawberry
- 11 cake tops were supplied via Source_CakeTop

```
-- 1. Clear previous data from older simulation runs
ProductionReport.delete;

-- 2. Log Raw Parts produced by Machining Stations
ProductionReport[1, 1] := "Total Cylinders Processed";
ProductionReport[2, 1] := Lathe_L1.statNumOut;

ProductionReport[1, 2] := "Total V-Shapes Processed";
ProductionReport[2, 2] := Milling_L2.statNumOut;

-- 3. Log Secondary Parts produced by independent sources
ProductionReport[1, 3] := "Total Hooks Supplied";
ProductionReport[2, 3] := Source_Strawberry.statNumOut;

ProductionReport[1, 4] := "Total Cake Tops Supplied";
ProductionReport[2, 4] := Source_CakeTop.statNumOut;

-- 4. Log Production at Assembly Stages
ProductionReport[1, 5] := "Assemblies Built (Assembly_CIMA)";
ProductionReport[2, 5] := AssemblyStation1.statNumOut;

ProductionReport[1, 6] := "Final Cakes Built (Assembly_Cake)";
ProductionReport[2, 6] := AssemblyStation2.statNumOut;

-- 5. Log Shipped Items reaching the Exit Drain
ProductionReport[1, 7] := "Final Packed Boxes (Drain)";
ProductionReport[2, 7] := Drain1.statNumOut;
```

Figure 49. GenerateFinalReport Method

Total Cylinders Processed		
	string 1	integer 2
1	Total Cylinders Processed	10
2	Total V-Shapes Processed	20
3	Total Hooks Supplied	11
4	Total Cake Tops Supplied	11
5	Assemblies Built (Assembly_CIMA)	10
6	Final Cakes Built (Assembly_Cake)	10
7	Final Packed Boxes (Drain)	10
8		

Figure 50. Production Report Table

- **Assembly Efficiency:** The assembly stations processed all 10 sub-assemblies from AssemblyStation1 (1 cylinder + 2 V-shapes) and successfully combined them with hooks and cake tops at AssemblyStation2 to create 10 final products. The slight over-supply of hooks and cake tops (11 units each) provided a buffer to prevent assembly line stoppages.
- **Time Distribution Analysis:** The final Summary Report reveals that:
 - 75.57% of time was spent on production activities (machining, assembly, polishing, inspection, packing)
 - 24.43% of time was consumed by transport and material handling operations
 - 17.74% value added represents the actual transformation time that directly contributes to product value

- 0.00% storage time indicates an efficient flow with minimal waiting in buffers

.Models.Model								
Simulation time:1:00:00:00.0000								
Object	Name	Mean Life Time	Throughput	TPH	Production	Transport	Storage	Value added
Drain1	Box	3:45.4703	10	0	75.57%	24.43%	0.00%	17.74%

Cumulated Statistics of the Parts which the Drain Deleted

Figure 51. Final Output Metrics

These results validate that the simulation accurately represents a real manufacturing environment with proper material flow, realistic processing times, and efficient use of resources while meeting all the challenge requirements.

E. Team Innovation Module

1. **Segmented access:** To address security issues within the manufacturing cell, it was decided to add an authentication stage to the cell based on an RFID sensor. Whenever an user wants to get access to the cell's functions, he/she will have to present his identification card to the sensor and depending on his/her assigned role, different interfaces will be or not accessible.

For this purpose a completely external system was designed. To make it more robust, a PCB was designed for the electronic system. It mainly consists of four relays that are activated in different patterns depending on the code of each user's credential; these relays are wired to four digital inputs of the PLC and through the programming all the functions are established.

Additionally, an LCD screen to display important information was added and a case to encapsulate all the electronic components, the PCB and the wirings was designed.



Figure 52. 3D RFID System

2. **Bigger number of available classifications:** As previously mentioned, contrary to the original requirements, the vision sensor and the PLC are capable to detect three different classifications, and perform three different processes instead of only two; this was achieved by modifying certain

parameters of the Cognex sensor and adding extra steps to the ladder logic in order to detect invalid classifications.

3. **The output of the cell is different to the input:** To simulate a fully terminated item, the cell is capable of physically assembling a product that is different to the original input piece. Taking advantage of the team's proposal to simulate a cake factory, the input to the cell is a piece that simulates the base (first floor) of a cake, which, after completing its travel through the cell, is attached to a smaller piece (that simulates the second floor) with the use of the cobot. Therefore, the output of the cell is a fully terminated cake. It is important to note that the top of the cake corresponds to its base, so if a chocolate base is the input to the system, a full chocolate cake will be the output, and the same with the vanilla and strawberry flavors.



Figure 53. 3D Fully Assembled Cakes (Strawberry, Chocolate and Vanilla)

4. **Integrated dashboard:** The main dashboard is a compendium of the most important parameters of both the cell and the lathe located at CIMA, achieving a full integration of both process metrics for a general view of the complete industrial system's performance. Besides, two separated dashboards integrated into the same web address, allow the access to a wider view of each stage of the production process where more specific metrics for the cell and the lathe can be found separately. Additionally, taking advantage of Tec's internet, the dashboard can be accessed from any device connected to its local campus network.

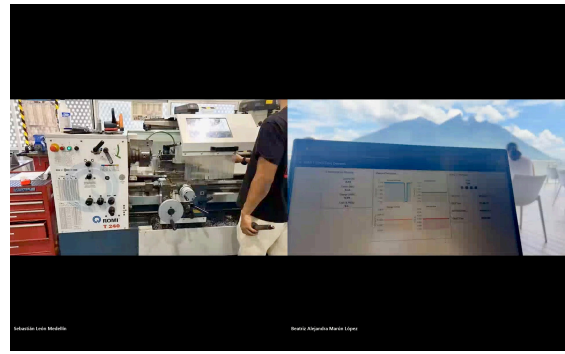
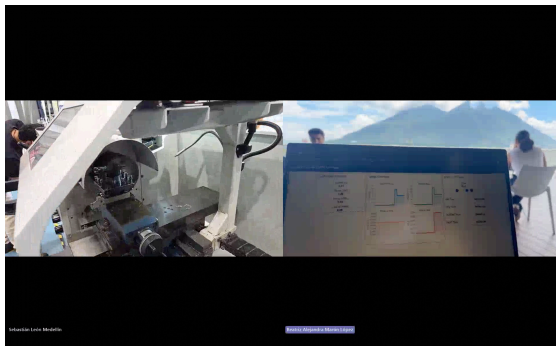


Figure 54 & 55. Real Time Lathe Monitoring on Monterrey Campus

5. **Power, Energy, and Cost Estimation:** Beyond raw current measurement, the IoT monitoring system was extended to estimate a comprehensive set of electrical performance indicators in real time. Using the measured RMS current, an assumed line voltage of 220 V, and a representative power factor of 0.85, the system calculates the estimated three-phase active power demand. Energy consumption is then obtained by integrating power over time, and an operating cost estimate is continuously updated using a reference electricity rate of 2.50 MXN/kWh based on CFE tariff data. These indicators are published via MQTT and displayed on the Node-RED dashboard, providing operators with actionable economic metrics alongside operational data.
6. **FIFO-Based Per-Piece Process Tracking:** To handle simultaneous requests of different flavors on the line without mixing up their associated processing times, a FIFO (First In, First Out) indexing strategy was implemented in the PLC. Dedicated data blocks were created for drilling, and mixing processing stations, as well as for the cobot tracking logic. Every time the vision sensor classifies a new piece, its flavor ID is stored in the next available position of each array. A separate read index advances independently as each piece reaches its corresponding station, ensuring that the multiplexor always applies the correct processing time for the piece currently being operated on, regardless of how many pieces are on the line simultaneously. This architecture guarantees process integrity even under continuous, multi-flavor production conditions.
7. **Enhanced palletizing:** To simulate a ready-to-sell product, a structured two-level palletizing sequence was designed and implemented. The palletizing cage consists of three physical components: a base tray, an intermediate separator lid, and a top cover lid. Each level accommodates a 2x2 arrangement of fully assembled cakes, for a total capacity of 8 units per palletizing cycle. The cobot places the first four finished cakes onto the lower level of the base tray. Once four units are

placed, it picks and positions the intermediate separator lid on top of the first layer, and subsequently fills the upper level with the next four cakes. Finally, the top cover lid is placed to seal the pallet. Upon completion of the 8-unit cycle, all counters and positional indices are automatically reset, allowing the system to assume a new empty cage is in place and repeat the sequence continuously without operator intervention.

8. **Custom End-Effectors and Cell Fixtures:** A custom gripper was developed for the xArm Lite 6 cobot, capable of securely grasping cake bases, second-floor topping pieces, and palletizing separator lids. A pneumatic funnel was designed to guide the cobot's placement of each cake base onto the cell's entry point, ensuring consistent and repeatable positioning at the start of the pneumatic process and reducing the risk of misalignment.

Two 3×3 supply grids were fabricated to organize and store the raw cake bases and topping pieces in fixed, indexed positions. These grids allow the cobot to navigate to each slot using precalculated coordinates, supporting the per-flavor pick sequence without the need for additional sensing.

9. **Analog Signal Integration:** To extend the cell's sensing capabilities beyond discrete digital signals, pressure monitoring is acquired through the PLC's analog input module and transmitted to the Node-RED dashboard via the MQTT infrastructure, where it is displayed as a real-time pressure indicator in Pascals. This addition provides operators with continuous visibility of the pneumatic system's operating condition, enabling early detection of pressure drops or abnormal values that could indicate leaks, actuator faults, or supply irregularities, contributing to more proactive maintenance and safer cell operation.

III. RESULTS AND DISCUSSION

a. System Data Analysis

i. Analysis of Data Acquired Through the Node-RED Dashboard

The results presented in the following analysis were obtained through the instrumentation of a ROMI T240 lathe using an SCT current sensor for electrical current monitoring. The acquired measurements were transmitted via the MQTT protocol to a Node-RED dashboard, where the data were stored both in a CSV file and in a SQL database for subsequent processing and analysis.

During the monitored period, the **machine operated for a total of 1.80 hours** (107.95 minutes). Based on the measured current values, the machine operating condition was classified into four discrete states using predefined current thresholds. The classification logic implemented in the data analysis was defined as follows:

- **OFF:** Current < 0.40 A
- **IDLE:** 0.40 A ≤ Current ≤ 7.30 A
- **CUTTING:** 7.30 A < Current ≤ 15.00 A

- **FAULT:** Current > 15.00 A

The OFF threshold was established to identify periods during which the machine spindle and drive systems were not drawing significant electrical power. Current values below 0.40 A were primarily associated with machine inactivity, part inspection, dimensional verification using measuring instruments, process planning activities, and tool entry and exit movements that did not involve material removal.

The IDLE state corresponds to machine operation without active cutting, including spindle preparation, positioning movements, setup adjustments, and short waiting periods between machining operations. The upper threshold of 7.30 A was determined experimentally from the observed current profile, representing the transition between non-productive operation and material removal.

The CUTTING state was defined for current values above 7.30 A and below 15.00 A. This range was associated with active machining operations, during which the spindle motor and feed mechanisms experienced increased electrical demand due to the cutting forces generated during turning. The threshold was selected based on the measured current distribution and visual correlation with the machining process.

Finally, current values exceeding 15.00 A were classified as FAULT conditions within the monitoring algorithm. However, these events did not necessarily correspond to actual machine faults. During the experimental evaluation, most occurrences above this threshold were associated with the transient inrush current generated during machine startup. When the ROMI T240 lathe was powered on, a short-duration current peak was observed as the electrical and drive systems were energized. Consequently, the FAULT classification should be interpreted as a high-current condition rather than a confirmed machine failure. The threshold was retained to identify abnormal current peaks and facilitate the detection of potential overloads or unexpected operating conditions during future deployments.

As shown in Table 7, the OFF state accounted for 63% of the monitored period. Although this percentage may initially appear high, a significant portion of this time was dedicated to non-machining activities required for part production, including dimensional measurements, diameter verification, machining calculations, process planning, and tool approach and withdrawal operations. Consequently, the OFF state should not be interpreted exclusively as machine downtime, but rather as a combination of inactive and auxiliary production activities necessary for manufacturing the workpiece.

State	Percentage	Time (h)
OFF	63%	1.133333333
IDLE	3%	0.062777778

CUTTING	33%	0.593888889
FAULT	1%	0.009166667

Table 7. Operating Time Distribution by Machine State

The recorded **electrical current ranged from 0.01 A to 15.18 A**, with an **average value of 3.25 A**.

The power profile obtained from the electrical measurements reflects the different operating conditions of the machine throughout the monitored period. The **average power demand during the CUTTING state was higher than during the OFF and IDLE states**, as expected due to the energy required to perform the machining operation. The **maximum recorded power reached 4.92 kW**, while the **average power throughout the monitoring period was 1.05 kW**. The calculated **load factor of 21%** indicates that the machine operated at a relatively low average demand compared to its peak power requirement.

A **total energy consumption of 1.70 kWh** was measured during the analyzed period. The energy distribution among the identified machine states is presented in Table 8.

State	Energy consumption	Percentage
OFF	0.09 kWh	5%
IDLE	0.13 kWh	8%
CUTTING	1.43 kWh	84%
FAULT	0.04 kWh	3%

Table 8. Energy Consumption By Machine State

The CUTTING state accounted for the largest share of energy consumption, representing 84% of the total measured energy. The remaining **16%** was associated with the OFF, IDLE, and FAULT states (**non-productive states**), which together consumed 0.26 kWh.

To characterize the operational behavior of the machine, a state transition analysis was performed. Table 9 summarizes the detected transitions and their corresponding interpretations.

Transition	Events	Meaning
OFF to IDLE	4	Machine start-up Frequent shutdowns in the machining
IDLE to CUTTING	27	Production start-up Constant restarts in the process
CUTTING to IDLE	37	Normal shutdown Microstops in the process

IDLE to OFF	14	Total shutdowns
CUTTING to FAULT	2	Incoming faults
FAULT to IDLE	0	Recovery faults

Table 9. State Transition Analysis

The **estimated electrical energy cost** for the analyzed manufacturing cycle was approximately **\$4.24 MXN** and it was calculated using an assumed energy rate of 2.50 MXN/kWh (CFE, 2026). This value was adopted as a representative electricity cost for comparative analysis and does not correspond to the complete billing structure of the utility provider. Therefore, fixed charges, demand charges, and other tariff components were excluded from the calculation.

The acquired dataset serves as the basis for the calculation of key performance indicators (KPIs) used to evaluate machine utilization, energy efficiency, operational availability, and process productivity. These indicators are further analyzed in the subsequent section of this report.

Overall, the results provide a quantitative characterization of the machine's operational behavior during the manufacturing of a single workpiece based on the electrical measurements obtained through the proposed IoT monitoring architecture. For additional details regarding the acquired data, data processing methodology, and analytical calculations, refer to the ROMI_Piece_0601_DataAnalysis Excel file available in the project GitHub repository.

ii. Manufacturing Metrics Calculations

This section presents the evaluation of key manufacturing performance metrics derived from the monitored machining process of a single workpiece on the ROMI T240 lathe. The analysis includes the cycle time, processing time, average production time, production rate, and production capacity, providing a quantitative characterization of the machining performance under job production conditions.

Assuming a batch of 100 parts and 40 work hours per week:

Metrics	Procedure
Cycle Time	$T_c = T_o + T_h + T_t$ Where: $T_o = \text{Assembly operation time } (\frac{\text{min}}{\text{pc}})$ $T_h = \text{Handling time of the workpiece } (\frac{\text{min}}{\text{pc}})$ $T_t = \text{Average tool handling } (\frac{\text{min}}{\text{pc}})$ From the data acquired: $T_o = 35.633 \frac{\text{min}}{\text{pc}} \text{ (CUTTING)}$ $T_h = 5.8 \frac{\text{min}}{\text{pc}}$ $T_t = 9.5 \frac{\text{min}}{\text{pc}}$

	$T_c = T_o + T_h + T_t$ $T_c = 35.633 \frac{\text{min}}{\text{pc}} + 5.8 \frac{\text{min}}{\text{pc}} + 9.5 \frac{\text{min}}{\text{pc}}$ $T_c = 50.933 \frac{\text{min}}{\text{pc}}$	From the data acquired: $T_p = 51.047 \frac{\text{min}}{\text{pc}}$ $R_p = \frac{60}{T_p}$ $R_p = \frac{60}{51.047 \frac{\text{min}}{\text{pc}}}$ $R_p = 1.175 \frac{\text{pc}}{\text{hr}}$
Batch Production Processing Time	$T_b = T_{su} + QT_c$ Where: $T_{su} = \text{Setup time } (\frac{\text{min}}{\text{pc}})$ $Q = \text{Work units}$ $T_c = \text{Cycle time } (\frac{\text{min}}{\text{pc}})$ From the data acquired: $T_{su} = 11.38 \frac{\text{min}}{\text{pc}}$ $Q = 100$ $T_c = 50.933 \frac{\text{min}}{\text{pc}}$ $T_b = T_{su} + QT_c$ $T_b = 11.38 \frac{\text{min}}{\text{pc}} + (100) 50.933 \frac{\text{min}}{\text{pc}}$ $T_b = 5104.68 \frac{\text{min}}{\text{batch}} = 85.078 \frac{\text{hr}}{\text{batch}}$	Production Capacity $C_p = nH_{pc} R_p$ Where: $n = \text{Number of machines}$ $H_{pc} = \text{Plant capacity } (\frac{\text{hr}}{\text{week}})$ $R_p = \text{Production rate } (\frac{\text{pc}}{\text{hr}})$ From the data acquired: $n = 1$ $H_{pc} = 40 \frac{\text{hr}}{\text{week}}$ $R_p = 1.175 \frac{\text{pc}}{\text{hr}}$ $C_p = nH_{pc} R_p$ $C_p = (1)(40 \frac{\text{hr}}{\text{week}})(1.175 \frac{\text{pc}}{\text{hr}})$ $C_p = 47 \frac{\text{pc}}{\text{week}}$
Job Production (Q = 5) Average Production Time	$T_p = T_{su} + T_c$ Where: $T_{su} = \text{Setup time } (\frac{\text{min}}{\text{pc}})$ $T_c = \text{Cycle time } (\frac{\text{min}}{\text{pc}})$ From the data acquired: $T_{su} = 11.38 \frac{\text{min}}{\text{pc}}$ $T_c = 50.933 \frac{\text{min}}{\text{pc}}$ $T_p = T_{su} + T_c$ $T_p = 11.38 \frac{\text{min}}{\text{pc}} + 50.933 \frac{\text{min}}{\text{pc}}$ $T_p = 62.31 \frac{\text{min}}{\text{pc}} = 1.0385 \frac{\text{hr}}{\text{pc}}$	
Batch Production (Q = 100) Average Production Time	$T_p = \frac{T_b}{Q}$ Where: $T_b = \text{Processing time } (\frac{\text{min}}{\text{batch}})$ $Q = \text{Work units}$ From the data acquired: $T_b = 5104.68 \frac{\text{min}}{\text{batch}} = 85.078 \frac{\text{hr}}{\text{batch}}$ $Q = 100$ $T_p = \frac{T_b}{Q}$ $T_p = \frac{5104.68 \frac{\text{min}}{\text{batch}}}{100}$ $T_p = 51.047 \frac{\text{min}}{\text{pc}}$	The OEE (Overall Equipment Effectiveness) is calculated based on the electrical monitoring data acquired during the machining of a single workpiece on the ROMI T240 lathe. Since only one piece was fully monitored, the following analysis assumes that all five produced pieces shared equivalent machining conditions and cycle times to the monitored workpiece. Under this assumption, the total planned production time was scaled proportionally. $T_{\text{planned}} = 5 \times 107.95 = 539.75 \text{ min} \approx 540 \text{ min}$ For Availability, Downtime is estimated as the accumulated OFF and FAULT state times per piece, scaled to 5 pieces: $\text{Downtime} = 5 \times (0.64 \times 107.95) = 5 \times 69.09 = 345.45 \text{ min}$ Thus: $A = \frac{540 - 345.45}{540} = 36.02\%$ For Performance, the ideal cycle time is taken as the CUTTING state duration of the monitored piece, representing pure value-added machining time: $P = \frac{5 \times 35.633}{540} = 32.99\%$ Of the 5 pieces produced, 3 were confirmed to meet dimensional and quality specifications, yielding a Quality factor of: $Q = \frac{3}{5} = 60\%$ Thus, the OEE is calculated as:
Batch Production Rate	$R_p = \frac{60}{T_p}$ Where: $T_p = \text{Average production time } (\frac{\text{min}}{\text{pc}})$	

Table 10. Manufacturing Metrics Calculations

$$OEE = A \times P \times Q = 0.3602 \times 0.3299 \times 0.60 = 0.0712\%$$

$$OEE = 7.13\%$$

The resulting OEE of approximately 7% is low, which is expected given the context of manual job-shop machining of a single custom workpiece on a conventional lathe. The primary contributors to this result are:

- **Low Availability:** a large proportion of the monitored time corresponded to OFF states associated with dimensional verification, process planning, tool changes, and part inspection, all of which are inherent to manual machining operations
- **Low Performance:** the ideal cycle time represents only the pure cutting portion, while the full cycle includes significant handling and tool management time
- **Moderate Quality:** a 60% quality rate reflects the challenges of achieving tight dimensional tolerances on a manually operated conventional lathe

It is important to note that this OEE value should not be interpreted as representative of optimized production. Rather, it provides a baseline characterization of the machining process under job-shop conditions for a low-volume, high-complexity workpiece.

b. Cycle Times

To evaluate the end-to-end performance of the manufacturing system, cycle times were measured and mapped across every major stage of the production process, from raw material machining at CIMA through final palletizing at Cell 3105. Table 11 presents the breakdown in seconds for each stage, differentiating between product flavors where processing times vary.

Process stage	Cycle time (s)
Machining (ROMI T240)	6477
Transfer to Vision Area	36
Vision Inspection (Cognex)	5
PLC Classification & Recipe Selection	2
Cobot Routines	69
Processing Stations (drilling, mixing, loading piston, turntable, pressing and conveyors)	Strawberry: 50 Chocolate: 46 Vanilla: 48
Assembly	17
Final Palletizing	30
Total	Strawberry: 6686 Chocolate: 6682 Vanilla: 6684

Table 11. Cycle Times Breakdown

The dominant contributor to total cycle time is the machining of the aluminum workpiece on the ROMI T240 lathe at CIMA, which accounts for 6,477 seconds (approximately 1 hour and 48 minutes) and represents over 97% of the total cycle regardless of flavor. This figure encompasses the full sequence of facing, roughing, and finishing operations required to produce the stepped cylindrical geometry, and is consistent with the 107.95-minute monitoring period reported in the IoT analysis. Because this stage is performed manually on a conventional lathe by a single operator, it constitutes the system's primary bottleneck and the most significant opportunity for cycle time reduction in a scaled production scenario.

Once the machined workpiece arrives at Cell 3105, the process transitions to automated operation. Physical transfer to the Cognex inspection area takes 36 seconds, after which the vision sensor classifies the part in approximately 5 seconds. This inspection time includes image capture triggered by the one-second TP pulse from the PLC, internal Cognex processing, and digital output transmission to the controller. PLC classification and recipe selection, including storage of the flavor ID into the FIFO array, S7 network transmission to the Pneumatic PLC, and multiplexor configuration adds 2 seconds, making the total vision-to-recipe latency 7 seconds. This is negligible relative to total cycle time and confirms that the vision system introduces no meaningful throughput penalty.

Processing station times vary by flavor due to the multiplexor-controlled piston pressing durations configured through the HMI. Strawberry requires the longest processing time at 50 seconds, followed by Vanilla at 48 seconds and Chocolate at 46 seconds. These times aggregate the pneumatic funnel loading, turntable rotation, vertical piston pressing sequence, inter-conveyor transfers, drilling station dwell, and mixing station dwell. The 4-second spread across flavors is small in absolute terms but demonstrates the dynamic recipe configuration capability of the dual-PLC architecture functioning as intended.

Cobot routines across all phases total 69 seconds: base pick-and-place from the supply grid to the funnel, inter-conveyor transfer from Conveyor 5 to Conveyor 6, and topping assembly. This figure reflects the sum of all physical robot motion time across Phases 2 through 6 of the FSM lifecycle, excluding waiting states where the robot is polling for PLC acknowledgment or conveyor sensor activation. The assembly stage, during which the cobot places the topping piece onto the cake base at Conveyor 6, contributes 17 seconds independently and is listed separately given its distinct functional role as the value-adding differentiation step of the cell.

Final palletizing, which includes covering the motion from the cobot's safe position to product pickup, placement onto the pallet grid at the correct 2×2 slot, and return to safe position, requires 30 seconds per unit. Separator lid placement, which occurs after every 4 units, is not included in this per-unit figure as it occurs at a lower frequency and would skew per-cycle comparison.

The resulting total cycle times are 6,677 seconds for Strawberry, 6,673 seconds for Chocolate, and 6,675 seconds for Vanilla. The flavor-to-flavor variation of at most 4 seconds confirms that recipe differentiation has a negligible impact on overall throughput at the system level, and that any meaningful cycle time improvement must target the machining stage.

c. Automation Highlights

This section delineates the most robust, efficient, and technically refined elements of the implemented mechatronic system. The highlights span across collaborative robot task management, Programmable Logic Controller (PLC) sequencing, computer vision integration, and Industrial Internet of Things (IIoT) data acquisition.

i. Collaborative Robot

One of the most significant improvements was the implementation of a Finite State Machine (FSM) architecture to replace conventional sequential execution. Through this approach, each product is managed as an independent task entity, enabling the simultaneous processing of two products within the production pipeline without requiring multi-threaded robot motion control. This design introduces a structural throughput improvement that surpasses the performance gains achievable through trajectory optimization alone within a traditional linear execution model.

Task allocation and workflow coordination are managed through a priority-based scheduling mechanism designed to maintain continuous material flow throughout the cell. The scheduler prioritizes downstream operations before admitting new tasks into the system, thereby minimizing the possibility of bottlenecks or congestion within the production process. Additionally, active duplicate-task suppression mechanisms maintain synchronization between the virtual product registry and the physical inventory, ensuring consistency during continuous operation and preventing state mismatches that could compromise system reliability.

System robustness and operational safety were further enhanced through dedicated fault isolation strategies. Safety monitoring processes execute independently from the primary control loop through background threads, allowing emergency conditions to be handled deterministically and with minimal response latency. Furthermore, monotonic timeout mechanisms were incorporated into all blocking states, ensuring that sensor failures, communication interruptions, or missing PLC signals initiate controlled recovery procedures instead of causing indefinite system freezes.

ii. IIoT Integration

One of the most significant achievements of the project was the successful integration of an Industrial Internet of Things (IIoT) monitoring architecture into the manufacturing process. The developed system enabled real-time acquisition, processing, transmission, and visualization of electrical consumption data from the ROMI T240 lathe

using a low-cost ESP32 gateway and an SCT-013 current sensor.

A particularly robust aspect of the implementation was the development of a complete end-to-end monitoring solution, beginning with non-invasive current sensing, followed by signal conditioning, data acquisition through a 16-bit ADC, local processing on the ESP32, MQTT-based communication, and real-time visualization through Node-RED dashboards. The system continuously calculated current, power consumption, energy usage, operating cost, and machine operating state, providing valuable insights into machine utilization and process performance.

Another notable achievement was the integration of two independent data acquisition environments into a unified monitoring platform. Data generated by the IIoT gateway in Plant 1 and operational information acquired from the PLC in Cell 3105 were successfully consolidated through a common MQTT infrastructure. This architecture enabled centralized supervision of machining operations, production metrics, utility consumption, and equipment status from a single monitoring environment.

The developed dashboards provided remote accessibility through the Tecnológico de Monterrey network, allowing operators and supervisors to monitor the manufacturing systems from any campus across the country.

iii. RFID Interface

Access to both HMIs, and by extension to the entire manufacturing cell, is controlled through a custom-designed RFID authentication device. The system reads the operator's student ID card using a standard RFID reader paired with an ATmega328P microcontroller. Authentication is communicated to the PLC through four relays, each driving a 24 VDC digital input — forming a 4-bit signal that encodes the granted access level. The device is powered directly from the PLC's existing 24 VDC supply, requiring no additional power connections. All circuitry is mounted on a custom printed circuit board, ensuring long-term reliability in an industrial environment.

d. Challenges and Opportunities for Improvement

i. Collaborative Robot

During the initial development phase, the native UFACTORY Blockly interface was evaluated and deemed insufficient for the operational complexity of Cell 3105. Its single-threaded, sequential execution model lacked the mechanisms required to handle concurrent workpieces, multi-flavor logic, bidirectional PLC signaling, and dynamic palletizing patterns simultaneously. This limitation necessitated a complete reimplementation of the control logic in Python utilizing the native xArm-Python-SDK.

Physical testing revealed two principal sources of operational instability. The first was transient signal noise on the configurable inputs (CI), wherein rapid or bouncing PLC signals risked being interpreted as new production orders before the preceding task had advanced, effectively duplicating products within the virtual registry. This was mitigated through the implementation of a 50 ms software

sampling window for all flavor reads, coupled with an explicit duplicate-suppression algorithm that cross-references active tasks by flavor and tray index prior to registration.

The second challenge involved state desynchronization between the physical tray and the virtual registry; if a pick operation failed mid-sequence, the column index risked advancing regardless, causing the manipulator to target an empty slot in subsequent cycles. This was resolved by conditioning the increment of the `cake_count` variable exclusively upon confirmed, successful pick operations, thereby maintaining strict synchronization between physical and virtual states.

Furthermore, managing multiple concurrent products introduced the risk of resource contention. Without a defined execution hierarchy, the scheduler risked prioritizing easily serviceable tasks over critical ones, potentially leading to conveyor blockages and assembly station congestion. The implementation of a fixed-priority scheduling hierarchy resolved this by ensuring that downstream operations are systematically cleared first. In practice, this architecture successfully allowed up to two workpieces to coexist at distinct pipeline stages without deadlock or positional conflict.

ii. IoT Integration

Several technical challenges were encountered during the development and testing of the IoT monitoring system. One of the primary challenges encountered during the development of the IoT monitoring system was the proper conditioning and interpretation of the signal generated by the SCT-013 current sensor. Since the sensor produces a low-amplitude AC signal, it was necessary to design a dedicated signal conditioning circuit capable of adapting the waveform to the operating range of the ADS1115 analog-to-digital converter. Several iterations were performed to correctly implement the burden resistor, voltage divider network, filtering capacitors, and operational amplifiers configured as voltage followers. These modifications ensured a stable virtual ground, minimized signal distortion, and improved measurement reliability. In addition to the hardware design, considerable effort was devoted to developing and calibrating the acquisition algorithm, including the sampling strategy, peak detection method, and current calculation procedure, to obtain consistent and representative measurements of the lathe's electrical consumption.

Another challenge was associated with the estimation of electrical power and energy consumption. Since only current measurements were available, power calculations relied on assumed values for line voltage and power factor. Although this approach provided useful operational insights, future implementations could incorporate direct voltage measurement and true power calculation techniques to improve overall measurement accuracy.

Communication reliability also represented an important consideration. Because the system depended on wireless communication and MQTT message exchange, temporary

network interruptions could affect data availability. To address this issue, reconnection routines were implemented within the ESP32 firmware, allowing automatic recovery of Wi-Fi and MQTT connections whenever communication losses occurred.

From the monitoring perspective, the machine-state classification algorithm successfully identified OFF, IDLE, CUTTING, and high-current conditions. However, some events classified as FAULT states were later identified as normal startup current transients rather than actual machine faults. Future versions could incorporate adaptive thresholds, or machine-learning-based classification techniques to improve the accuracy of state detection.

The dashboard infrastructure also offers several opportunities for future expansion. Potential improvements include automated report generation, cloud-based data storage, predictive maintenance indicators, energy-efficiency analytics or alarm notification systems. Additionally, incorporating historical trend analysis and machine performance benchmarking could further enhance decision-making capabilities.

IV. CONCLUSIONS

The development and physical implementation of the intelligent pastry manufacturing cell demonstrated that the integration of industrial-grade automation technologies with low-cost embedded IoT systems is not only technically feasible, but capable of producing a cohesive, production-ready solution that meets and exceeds the functional requirements of a real manufacturing environment.

Working with Siemens S7-1200 PLCs, a Cognex industrial vision sensor, and a UFACTORY xArm Lite 6 collaborative robot provided the team direct exposure to the communication protocols, timing constraints, and signal integrity considerations that define real industrial automation practice. The experience of establishing bidirectional S7 networks, manage discrete I/O synchronization between heterogeneous devices, and debugging ladder logic under physical operating conditions reinforced the gap that exists between theoretical automation knowledge and its practical application; a gap that can only be bridged through hands-on implementation.

The decision to implement the collaborative robot's control architecture in Python rather than the native Blockly interface proved to be one of the most technically significant choices of the project. It required a deeper understanding of concurrent task management, state machine design, and hardware-level communication, and ultimately produced a system capable of behaviors that would have been structurally impossible within a linear programming model, like priority-based scheduling, pipeline parallelism, and fault-tolerant shutdown. This reinforces the lesson that selecting the right level of abstraction for a control problem is as critical as the solution itself.

The IoT instrumentation developed for the ROMI T240 lathe illustrated how meaningful process intelligence can be extracted from industrial machinery using accessible, low-cost components, provided that the signal conditioning, communication architecture, and data processing pipeline are designed with sufficient rigor. The end-to-end system, from current sensing through MQTT transmission to real-time dashboard visualization, demonstrated that the boundary between embedded electronics and industrial monitoring is increasingly defined by software and systems thinking rather than hardware cost.

The Tecnomatix Plant Simulation model complemented the physical implementation by providing a structured framework for analyzing material flow, resource utilization, and production throughput at a system level. The ability to validate cycle times, identify bottlenecks, and quantify time distribution across productive and non-productive states before committing to physical changes is a capability with direct relevance to industrial engineering practice.

Taken together, the project established that modern manufacturing systems are fundamentally interdisciplinary: their performance depends equally on mechanical design, electrical engineering, control logic, software architecture, data infrastructure, and human factors such as access control and operator interface design. The team's experience spanning all of these domains within a single integrated system constitutes the most valuable outcome of this work: the practical understanding that in industry, these disciplines do not operate in isolation, and that the engineer capable of reasoning across all of them is the one capable of building systems that actually work.

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VI. PROJECT FILES REPOSITORY

<https://github.com/sLeon2005/Team-ATOLE-Manufacturing-Cell>